Receiver POVM (receiver)

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Every definition, fact, assumption, result and requirement of the contract, with the contract's numbers; remarks, examples, technical steps and all proofs are left out. Each heading opens its statement in the contract.

Mathematics

1 Modes, Fock space and the mode algebra

Definition 1.1 (Declared mode space)**.** A *declared mode space* is a finite totally ordered set $\mathbb{M}$, each of whose elements is an *optical mode*, together with the Hilbert space $\mathbb{C}^{\mathbb{M}}$ carrying the orthonormal basis $\{|m\rangle\}_{m\in\mathbb{M}}$ indexed in the declared order. A mode is a label $m = (p, \pi, \tau)$ recording a spatial path p, a polarization $\pi \in \{H, V\}$ and a temporal/spectral index τ; the order is lexicographic in (p, τ, π) with H before V. The index τ is *reserved*: it is carried in the label and in the emitted artifact, it may take only a single declared value in this contract, and no catalogue entry (Definition 16.2) reads or writes it. For $\Sigma \subseteq \mathbb{M}$ we write $E_\Sigma = \sum_{m\in\Sigma} |m\rangle\langle m|$ for the coordinate projector, and $\mathbb{M}$ is identified with $\{0, \dots, |\mathbb{M}| - 1\}$ when matrix indices are needed.

Definition 1.2 (Fock space and the second-quantization functor)**.** For a finite-dimensional Hilbert space $\mathcal{K}_0$ the *Fock space* is $\mathcal{F}(\mathcal{K}_0) = \bigoplus_{n\geq 0} \mathrm{Sym}^n(\mathcal{K}_0)$, where $\mathrm{Sym}^n(\mathcal{K}_0) \subseteq \mathcal{K}_0^{\otimes n}$ is the subspace invariant under every permutation of the n tensor factors and $\mathrm{Sym}^0(\mathcal{K}_0) = \mathbb{C}\,|\mathrm{vac}\rangle$. For a linear operator X on $\mathcal{K}_0$ put

$$\mathfrak{F}(X) \coloneqq \bigoplus_{n\geq 0} \left(X^{\otimes n}\right)\big|_{\mathrm{Sym}^n}, \qquad \mathfrak{F}(X)\big|_{\mathrm{Sym}^0} = 1. \tag{1.1}$$

This is well defined because $X^{\otimes n}$ commutes with every permutation of the factors and therefore preserves $\mathrm{Sym}^n(\mathcal{K}_0)$ (Lemma 1.6). The same formula defines $\mathfrak{F}(X) : \mathcal{F}(\mathcal{K}_0) \to \mathcal{F}(\mathcal{K}_1)$ for a linear map $X : \mathcal{K}_0 \to \mathcal{K}_1$ between two spaces, because $X^{\otimes n}$ intertwines the permutation actions and therefore carries $\mathrm{Sym}^n(\mathcal{K}_0)$ into $\mathrm{Sym}^n(\mathcal{K}_1)$. Occupation vectors $\boldsymbol{m} \in \mathbb{N}_0^{\dim \mathcal{K}_0}$ with $|\boldsymbol{m}| = n$ index the orthonormal basis $\{|\boldsymbol{m}\rangle\}$ of $\mathrm{Sym}^n(\mathcal{K}_0)$ obtained by symmetrizing and normalizing $|e_1\rangle^{\otimes m_1} \otimes \cdots$ in a fixed orthonormal basis $\{|e_j\rangle\}$ of $\mathcal{K}_0$; a_j and $a_j^\dagger$ denote the annihilation and creation operators of $|e_j\rangle$.

Definition 1.3 (Local element and its embedding)**.** A *local element* on a mode subset $\Sigma \subseteq \mathbb{M}$ is a matrix $G_\Sigma \in \mathbb{C}^{\Sigma\times\Sigma}$ with $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}_\Sigma$; Σ is its

support. Its *global embedding* is

$$\hat{G} \coloneqq G_\Sigma \oplus \mathbb{1}_{\mathbb{M}\setminus\Sigma} \;\in\; \mathbb{C}^{\mathbb{M}\times\mathbb{M}}, \tag{1.2}$$

that is, $\hat{G}\,E_\Sigma = G_\Sigma E_\Sigma$ and $\hat{G}\,(\mathbb{1} - E_\Sigma) = \mathbb{1} - E_\Sigma$. A local element with $G_\Sigma^\dagger G_\Sigma = \mathbb{1}_\Sigma$ is a *lossless element*.

Proposition 1.4 (Algebra of embedded local elements)**.** Let $\hat{G}, \hat{G}'$ be global embeddings of local elements with supports Σ, Σ'.

(i) *Embedding preserves contractivity:* $\hat{G}^\dagger\hat{G} \preceq \mathbb{1}_\mathbb{M}$, with equality exactly when the element is lossless.

(ii) *Products of contractions are contractions:* if $X^\dagger X \preceq \mathbb{1}$ and $Y^\dagger Y \preceq \mathbb{1}$ then $(XY)^\dagger(XY) \preceq \mathbb{1}$, with equality for all vectors exactly when Y is an isometry and X is isometric on $\operatorname{ran} Y$.

(iii) *Disjoint supports commute:* if $\Sigma \cap \Sigma' = \varnothing$ then $\hat{G}\hat{G}' = \hat{G}'\hat{G}$.

(iv) *Adjoints and unitaries:* $\hat{G}^\dagger$ is the embedding of $G_\Sigma^\dagger$, and $\hat{G}$ is unitary exactly when G_Σ is.

Lemma 1.6 (Functoriality of the Fock lift)**.** Let $\mathcal{K}_0$ be finite-dimensional and let X, Y be operators on $\mathcal{K}_0$. Then, with $\mathfrak{F}$ as in (1.1):

(i) $X^{\otimes n}$ preserves $\operatorname{Sym}^n(\mathcal{K}_0)$, so $\mathfrak{F}(X)$ is well defined, and it is block diagonal with respect to the photon-number decomposition: $\mathfrak{F}(X)P_n = P_n\mathfrak{F}(X)$ for every n;

(ii) $\mathfrak{F}(XY) = \mathfrak{F}(X)\mathfrak{F}(Y)$, $\mathfrak{F}(\mathbb{1}) = \mathbb{1}$, and $\mathfrak{F}(X)^\dagger = \mathfrak{F}(X^\dagger)$;

(iii) if U is unitary then $\mathfrak{F}(U)$ is unitary; if $X^\dagger X \preceq \mathbb{1}$ then $\|\mathfrak{F}(X)\| \le 1$, so $\mathfrak{F}(X)$ is a bounded operator on $\mathcal{F}(\mathcal{K}_0)$;

(iv) if P is an orthogonal projector then $\mathfrak{F}(P)$ is the orthogonal projector onto $\mathcal{F}(\operatorname{ran} P) \subseteq \mathcal{F}(\mathcal{K}_0)$ — equivalently, with the occupation vectors formed in an orthonormal basis of $\mathcal{K}_0$ adapted to $\operatorname{ran} P \oplus \ker P$, onto the closed span of those carrying no photon outside $\operatorname{ran} P$;

(v) if U is unitary then $\mathfrak{F}(U)$ implements the mode transformation on the annihilation operators, $\mathfrak{F}(U)^\dagger a_i \mathfrak{F}(U) = \sum_j U_{ij} a_j$.

Lemma 1.9 (Spectrum of a lifted no-click operator on a photon-number block)**.** Let A be Hermitian on $\mathcal{K}_0$ with $0 \preceq A \preceq \mathbb{1}$ and eigenvalues $\lambda_1, \dots, \lambda_d$ ($d = \dim \mathcal{K}_0$). Then $\mathfrak{F}(\mathbb{1} - A)$ is a positive semidefinite contraction, its n-photon block $\mathfrak{F}(\mathbb{1} - A)^{(n)} = \big((\mathbb{1} - A)^{\otimes n}\big)\big|_{\operatorname{Sym}^n}$ has spectrum

$$\Big\{ \textstyle\prod_{j=1}^d (1-\lambda_j)^{m_j} \;:\; \boldsymbol{m} \in \mathbb{N}_0^d,\ |\boldsymbol{m}| = n \Big\}, \tag{1.4}$$

and consequently

$$\big(1-\lambda_{\max}(A)\big)^n\,\mathbb{1} \;\preceq\; \mathfrak{F}(\mathbb{1}-A)^{(n)} \;\preceq\; \big(1-\lambda_{\min}(A)\big)^n\,\mathbb{1}, \tag{1.5}$$

both bounds being attained as eigenvalues.

Lemma 1.10 (Order monotonicity of the Fock lift)**.** If $X \succeq Y \succeq 0$ on $\mathcal{K}_0$ then $\mathfrak{F}(X) \succeq \mathfrak{F}(Y)$. Consequently, if $0 \preceq A \preceq A' \preceq \mathbb{1}$ then $\mathfrak{F}(\mathbb{1}-A) \succeq \mathfrak{F}(\mathbb{1}-A')$.

Proposition 1.11 (Normal-ordered form of a lifted no-click operator)**.** For $0 \preceq A \preceq \mathbb{1}$ on $\mathcal{K}_0$,

$$\mathfrak{F}(\mathbb{1}-A) = :\exp\!\big(-\boldsymbol{a}^\dagger A \boldsymbol{a}\big): \;\coloneqq\; \sum_{p\geq 0} \frac{(-1)^p}{p!} \sum_{i_1\ldots i_p} \sum_{j_1\ldots j_p} \Big(\prod_{r=1}^{p} A_{i_r j_r}\Big) a_{i_1}^\dagger \cdots a_{i_p}^\dagger a_{j_1} \cdots a_{j_p}, \tag{1.7}$$

the series acting on each photon-number block as a finite sum.

2 The receiver instrument and its physical model

Definition 2.1 (Receiver instrument)**.** A *receiver instrument* is a tuple

$$\mathfrak{R} = \Big(\mathbb{M};\; \mathbb{I} = \mathbb{I}_C \sqcup \mathbb{I}_P;\; \{\mathbb{P}_k\}_{k=1}^{K};\; G\Big) \tag{2.1}$$

in which

(i) $\mathbb{M}$ is a declared mode space (Definition 1.1);

(ii) $\mathbb{I} \subseteq \mathbb{M}$ is the set of *declared input modes*, split into the *channel modes* $\mathbb{I}_C \neq \varnothing$ (the adversary's access) and the *declared source modes* $\mathbb{I}_P$ (the probe); every mode of $\mathbb{M} \setminus \mathbb{I}$ carries vacuum at the input (Assumption 11.4);

(iii) $\mathbb{P}_1, \ldots, \mathbb{P}_K$ are pairwise disjoint subsets of $\mathbb{M}$, the *detector ports*, with $K \geq 1$; a mode in no port is undetected;

(iv) $G \in \mathbb{C}^{\mathbb{M}\times\mathbb{M}}$ is the *instrument matrix*, indexed rows-by-outputs and columns-by-inputs, with $G^\dagger G \preceq \mathbb{1}$.

The basis $\{|m\rangle\}_{m\in\mathbb{I}}$ of $\mathbb{C}^{\mathbb{I}}$ is ordered with the channel modes $\mathbb{I}_C$ before the declared source modes $\mathbb{I}_P$, each part in the declared order of $\mathbb{M}$; matrices on $\mathbb{C}^{\mathbb{I}}$ and the columns of a matrix with column set $\mathbb{I}$ are indexed in this order, which differs from the declared order of $\mathbb{M}$ exactly when some declared source mode precedes some channel mode in it. We write $G_{\mathbb{I}} \in \mathbb{C}^{\mathbb{M}\times\mathbb{I}}$ for G restricted to the columns $\mathbb{I}$, and $E_J \coloneqq E_{\bigcup_{k\in J}\mathbb{P}_k}$ for $J \subseteq [K]$, where $[K] \coloneqq \{1, \ldots, K\}$. The channel-input system of the shared notation is the Fock space over the channel modes: $\mathcal{H}(C) = \bigoplus_n \mathcal{H}(C_n)$ with $\mathcal{H}(C_n) = \mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}_C})$ and photon-number projectors P_n^C.

Definition 2.2 (Port Grams and the loss deficiency)**.** The *port Gram matrices* of $\mathfrak{R}$ are the operators on $\mathbb{C}^{\mathbb{I}}$

$$A_k \coloneqq G_{\mathbb{I}}^{\dagger} E_{\{k\}} G_{\mathbb{I}}, \qquad k = 1, \dots, K, \tag{2.2}$$

and for $J \subseteq [K]$ we write $A_J \coloneqq \sum_{k \in J} A_k$ and $A_\Sigma \coloneqq A_{[K]}$. The *loss deficiency* is

$$\mathbb{L} \coloneqq \mathbb{1}_{\mathbb{I}} - A_\Sigma. \tag{2.3}$$

A family $\{A_k\}_{k=1}^K$ of operators on a finite-dimensional space is an *admissible Gram family* when $A_k \succeq 0$ for every k and $\sum_k A_k \preceq \mathbb{1}$. This is an exact condition; on a *declared* family it is *decided* in floating point at the receiver tolerance ε_2, by Definition 10.4 (i).

Definition 2.3 (Probe declaration)**.** A *probe declaration* is a pair $\big(\mathcal{T}_{\text{set}}, \{\rho^{(t)}\}_{t \in \mathcal{T}_{\text{set}}}\big)$ in which $\mathcal{T}_{\text{set}}$ is a finite non-empty alphabet of probe settings (including the setting off, whose state is vacuum) and, for each t, $\rho^{(t)}$ is a declared state on $\mathcal{F}(\mathbb{C}^{\mathbb{I}_P})$ that is *occupation-diagonal*: diagonal in the occupation basis of $\mathcal{F}(\mathbb{C}^{\mathbb{I}_P})$ (diagonal in the Fock basis). In the *Gaussian tier* each $\rho^{(t)}$ is a zero-mean Gaussian state with diagonal normally-ordered second-moment matrix $\Gamma^{(t)} = \operatorname{diag}(\bar{n}_m^{(t)})_{m \in \mathbb{I}_P} \succeq 0$, that is, a product of single-mode thermal states with mean photon numbers $\bar{n}_m^{(t)} \geq 0$; $\Gamma^{(t)} = 0$ is vacuum. A state of the Gaussian tier is occupation-diagonal by construction, a thermal state being diagonal in the occupation basis. When $\mathbb{I}_P = \varnothing$ the declaration is $\mathcal{T}_{\text{set}} = \{\text{off}\}$ and every statement about conditioning is vacuous.

Lemma 2.4 (Unitary dilation of a contraction)**.** Let G be a contraction on $\mathbb{C}^{\mathbb{M}}$ and let $\mathbb{E}$ be a disjoint copy of $\mathbb{M}$. With the defect operators $D_{\text{out}} \coloneqq (\mathbb{1} - GG^{\dagger})^{1/2}$ and $D_{\text{in}} \coloneqq (\mathbb{1} - G^{\dagger}G)^{1/2}$, the matrix

$$U_G = \begin{pmatrix} G & D_{\text{out}} \\ D_{\text{in}} & -G^{\dagger} \end{pmatrix} \quad \text{on} \quad \mathbb{C}^{\mathbb{M}} \oplus \mathbb{C}^{\mathbb{E}} \tag{2.4}$$

is unitary, and its block from the inputs $\mathbb{M}$ to the outputs $\mathbb{M}$ is G.

Definition 2.5 (The physical measurement model)**.** Fix a receiver instrument $\mathfrak{R}$ and a unitary extension U of G on $\mathbb{C}^{\mathbb{M} \sqcup \mathbb{E}}$ (Lemma 2.4 supplies one; any other may be used, and by Theorem 4.1 (v) the choice does not matter). Let $\iota : \mathbb{C}^{\mathbb{I}} \hookrightarrow \mathbb{C}^{\mathbb{M} \sqcup \mathbb{E}}$ be the inclusion of the declared input modes and $\mathfrak{F}(\iota) : \mathcal{F}(\mathbb{C}^{\mathbb{I}}) \to \mathcal{F}(\mathbb{C}^{\mathbb{M} \sqcup \mathbb{E}})$ the induced map, which under the factorization $\mathcal{F}(\mathbb{C}^{\mathbb{M} \sqcup \mathbb{E}}) \cong \mathcal{F}(\mathbb{C}^{\mathbb{I}}) \otimes \mathcal{F}(\mathbb{C}^{(\mathbb{M} \sqcup \mathbb{E}) \setminus \mathbb{I}})$, the occupation identification of Lemma 1.7 (ii), is $|\psi\rangle \mapsto |\psi\rangle \otimes |\text{vac}\rangle$: it prepares vacuum on every undeclared input mode and on every dilation mode (Assumption 11.4). The *measurement isometry* of the receiver is

$$V \coloneqq \mathfrak{F}(U)\, \mathfrak{F}(\iota) \;:\; \mathcal{F}(\mathbb{C}^{\mathbb{I}}) \longrightarrow \mathcal{F}(\mathbb{C}^{\mathbb{M} \sqcup \mathbb{E}}). \tag{2.7}$$

The detectors measure, on the output side, the K commuting binary projective measurements $\{\Pi_k^0, \mathbb{1}-\Pi_k^0\}$ where $\Pi_k^0 \coloneqq \mathfrak{F}(\mathbb{1}-E_{\{k\}})$, with the projector $E_{\{k\}}$ of $\mathbb{C}^{\mathbb{M}}$ extended by zero to $\mathbb{C}^{\mathbb{M}\sqcup\mathbb{E}}$, is the projector onto "no photon in the modes of $\mathbb{P}_k$" (Lemma 1.6 (iv), Assumption 11.7). For $J \subseteq [K]$ put $\Pi_J^0 \coloneqq \prod_{k\in J} \Pi_k^0$.

Lemma 2.6 (Exactness of vacuum elimination)**.** Let $S \subseteq T$ be finite sets of modes and $\iota : \mathbb{C}^S \hookrightarrow \mathbb{C}^T$ the coordinate inclusion. Under the factorization $\mathcal{F}(\mathbb{C}^T) \cong \mathcal{F}(\mathbb{C}^S) \otimes \mathcal{F}(\mathbb{C}^{T\setminus S})$ of occupation bases, the occupation identification of Lemma 1.7 (ii), $\mathfrak{F}(\iota)$ is $|\psi\rangle \mapsto |\psi\rangle \otimes |\mathrm{vac}\rangle$; and for any operator X on $\mathbb{C}^T$,

$$\mathfrak{F}(\iota)^\dagger\, \mathfrak{F}(X)\, \mathfrak{F}(\iota) \;=\; \mathfrak{F}\big(\iota^\dagger X \iota\big), \tag{2.8}$$

where $\iota^\dagger X \iota$ is the $S \times S$ block of X. In particular $\mathfrak{F}(\iota)$ is an isometry. The inclusion ι of Definition 2.5, with $S = \mathbb{I}$ and $T = \mathbb{M}\sqcup\mathbb{E}$, is one instance, and for it V of (2.7) is an isometry as well.

3 Component graphs and compilation

Definition 3.1 (Component catalogue)**.** A *component catalogue* $\mathfrak{C}$ is a finite set of entry types. An *entry type* fixes a support shape (which modes of $\mathbb{M}$ it acts on), a parameter domain, and a rule assigning to every parameter tuple in that domain a local element in the sense of Definition 1.3. The catalogue of this contract is the one of Definition 16.2 (Table 4) in the component-catalogue submodule, and Theorem 16.3 there states that every entry type maps its whole declared domain into local elements.

Definition 3.2 (Declared element graph)**.** A *declared element graph* is a finite directed graph $\mathfrak{E} = (V, E)$ whose vertices $e \in V$ are *elements*, each carrying a catalogue entry type, a parameter tuple, and a support $\Sigma_e \subseteq \mathbb{M}$. Write $\preceq_{\mathfrak{E}}$ for the reachability relation: $e \preceq_{\mathfrak{E}} f$ when $e = f$ or there is a directed path from e to f. The graph is *well formed* when

(E1) every element's parameter tuple lies in the declared domain of its entry type, so that, by Definition 3.1, G_{Σ_e} is a local element: defined, with $G_{\Sigma_e}^\dagger G_{\Sigma_e} \preceq \mathbb{1}$;

(E2) $\mathfrak{E}$ is acyclic, so that $\preceq_{\mathfrak{E}}$ is a partial order;

(E3) (*support comparability*) any two elements whose supports intersect are comparable: $\Sigma_e \cap \Sigma_f \neq \varnothing \implies e \preceq_{\mathfrak{E}} f$ or $f \preceq_{\mathfrak{E}} e$.

A *linear extension* of $\mathfrak{E}$ is an enumeration $e_1, \dots, e_L$ of V with $e_i \preceq_{\mathfrak{E}} e_j \implies i \leq j$.

Definition 3.3 (Compilation of a declared element graph)**.** Let $\mathfrak{E}$ be a well-formed graph; it is in particular acyclic (condition (E2) of Definition 3.2), so that a linear extension exists. For a linear extension $\sigma = (e_1, \dots, e_L)$ of $\mathfrak{E}$ put

$$G^{\sigma}_{\mathfrak{E}} \coloneqq \hat{G}_{e_L}\, \hat{G}_{e_{L-1}} \cdots \hat{G}_{e_1} \;\in\; \mathbb{C}^{\mathbb{M}\times\mathbb{M}}, \tag{3.1}$$

later elements multiplying from the left. The *compiled instrument matrix* of the receiver is $G \coloneqq D_\eta\, G^{\sigma}_{\mathfrak{E}}$, where $D_\eta = \sum_k \sqrt{\eta_k}\, E_{\{k\}} + E_{\mathbb{M}\setminus\bigcup_k \mathbb{P}_k}$ is the *terminal detector-efficiency layer*, applied after every element, and $\eta_k \in [0,1]$ is the declared efficiency of port k (the effect of that layer on the port Grams is Theorem 18.1 of the detector-imperfections submodule). The detector ports $\mathbb{P}_k$ and the port-embedding abbreviation $E_{\{k\}}$ written here are those of a receiver instrument tuple: clause (iii) of Definition 2.1 and its closing notation.

Theorem 3.4 (Well-definedness of compilation)**.** In items (i)–(iii), let $\mathfrak{E}$ be a well-formed declared element graph (Definition 3.2); in particular $\mathfrak{E}$ is acyclic (condition (E2) of Definition 3.2), so that $\preceq_{\mathfrak{E}}$ is a partial order and linear extensions of $\mathfrak{E}$ and of each of its induced subgraphs exist. Then:

(i) *Independence of the sort.* $G^{\sigma}_{\mathfrak{E}}$ is the same matrix for every linear extension σ of $\mathfrak{E}$; write $G_{\mathfrak{E}}$ for it. In particular, if V is partitioned into *layers* $L_1, \dots, L_r$ such that elements of one layer have pairwise disjoint supports and $e \preceq_{\mathfrak{E}} f$ implies $\mathrm{layer}(e) \le \mathrm{layer}(f)$, then $G_{\mathfrak{E}} = \big(\prod_{e\in L_r} \hat{G}_e\big) \cdots \big(\prod_{e\in L_1} \hat{G}_e\big)$ with each intra-layer product order-independent.

(ii) *Admissibility by construction.* $G^{\dagger}_{\mathfrak{E}} G_{\mathfrak{E}} \preceq \mathbb{1}$, and hence also $G^{\dagger} G \preceq \mathbb{1}$ for the compiled instrument matrix of Definition 3.3.

(iii) *Losslessness.* $G_{\mathfrak{E}}$ is unitary if and only if every element of $\mathfrak{E}$ is lossless on its support. If in addition every $\eta_k = 1$ then G is unitary.

(iv) *Decidability.* For an arbitrary declared element graph, not necessarily well formed, conditions (E1)–(E3) are decidable by finite inspection: (E1) by evaluating the declared domain predicate of each element, (E2) by a topological sort, (E3) by testing $\Sigma_e \cap \Sigma_f \neq \varnothing$ against the transitive closure of E for each of the $\binom{|V|}{2}$ pairs.

Corollary 3.5 (Receiver instrument from a well-formed graph declaration)**.** Let a route-(iii) declaration supply a declared mode space $\mathbb{M}$; a well-formed declared element graph $\mathfrak{E}$ on $\mathbb{M}$ (Definition 3.2), in particular acyclic (condition (E2) of Definition 3.2); a set $\mathbb{I} \subseteq \mathbb{M}$ of declared input modes with a split $\mathbb{I} = \mathbb{I}_C \sqcup \mathbb{I}_P$, $\mathbb{I}_C \neq \varnothing$, every mode of $\mathbb{M} \setminus \mathbb{I}$ carrying vacuum at the input (Assumption 11.4); and a port map $\mathbb{P}_1, \dots, \mathbb{P}_K$, pairwise disjoint subsets of $\mathbb{M}$ with $K \ge 1$, carrying detector efficiencies η_k in the domain $[0,1]$

declared for them in Definition 3.3. Let σ be a linear extension of $\mathfrak{E}$ and let $G \coloneqq D_\eta\, G^{\sigma}_{\mathfrak{E}}$ be the compiled instrument matrix of Definition 3.3. Then

(i) G does not depend on σ, so it is a function of the declaration alone; and

(ii) the tuple $\mathfrak{R} = \big(\mathbb{M};\ \mathbb{I} = \mathbb{I}_C \sqcup \mathbb{I}_P;\ \{\mathbb{P}_k\}_{k=1}^{K};\ G\big)$ is a receiver instrument in the sense of Definition 2.1: it has the shape (2.1) and satisfies its four clauses.

4 Port Grams and the click-pattern POVM

Theorem 4.1 (Port Grams and dilation independence)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}$ of Definition 2.2.

(i) *Admissibility.* $A_k \succeq 0$ for every k; $\sum_k A_k = G_{\mathbb{I}}^{\dagger} E_{[K]} G_{\mathbb{I}} \preceq G_{\mathbb{I}}^{\dagger} G_{\mathbb{I}} \preceq \mathbb{1}_{\mathbb{I}}$, so $\{A_k\}$ is admissible and the loss deficiency $\mathbb{L} = \mathbb{1}_{\mathbb{I}} - A_\Sigma$ is positive semidefinite.

(ii) *Rank.* $\operatorname{rank} A_k \le \min\big(|\mathbb{I}|, |\mathbb{P}_k|\big)$.

(iii) *Pullback.* For every $J \subseteq [K]$,

$$V^{\dagger}\, \Pi^{0}_{J}\, V = \mathfrak{F}\big(\mathbb{1}_{\mathbb{I}} - A_J\big). \tag{4.1}$$

(iv) *Vacuum elimination is exact.* The right-hand side of (4.1) depends on the instrument only through $G_{\mathbb{I}}$, the columns of G indexed by the declared input modes. Dropping the other columns is therefore an identity, not an approximation.

(v) *Dilation independence.* (4.1) holds for *every* unitary U on $\mathbb{C}^{\mathbb{M}\sqcup\mathbb{E}'}$, with any finite mode set $\mathbb{E}'$, whose block from the inputs $\mathbb{I}$ to the outputs $\bigcup_k \mathbb{P}_k$ equals the corresponding block of G. In particular the sub-unitary description (a contraction G on the declared modes) and the unitary-dilation description (a unitary on system plus environment, with vacuum environment and the environment discarded) induce the *same* measurement on $\mathcal{F}(\mathbb{C}^{\mathbb{I}})$.

Lemma 4.2 (Decomposition of the loss deficiency)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}$ and let E_{un} be the projector onto the declared modes lying in no port. Then

$$\mathbb{L} \;=\; \mathbb{1}_{\mathbb{I}} - A_\Sigma \;=\; \underbrace{\big(\mathbb{1}_{\mathbb{I}} - G_{\mathbb{I}}^{\dagger} G_{\mathbb{I}}\big)}_{\text{left through the dilation}} \;+\; \underbrace{G_{\mathbb{I}}^{\dagger}\, E_{\mathrm{un}}\, G_{\mathbb{I}}}_{\text{reached an unmonitored declared mode}}, \tag{4.3}$$

and both summands are positive semidefinite.

Definition 4.4 (Click-pattern alphabet)**.** An *outcome* is the set $S \subseteq [K]$ of ports that click; the *raw alphabet* is $2^{[K]}$, in bijection with $\{0,1\}^K$ by indicator vectors. We write

$$\mathsf{SC} \coloneqq \big\{\{k\} : k \in [K]\big\}, \qquad \mathsf{MC} \coloneqq \big\{S \subseteq [K] : |S| \geq 2\big\} \tag{4.4}$$

for the *single-click set* and the *multiclick set*, so that $2^{[K]} = \{\varnothing\} \sqcup \mathsf{SC} \sqcup \mathsf{MC}$, and $\varnothing$ is the *no-click outcome*.

Theorem 4.5 (Click-pattern POVM). Let $\mathfrak{R}$ be a receiver instrument with measurement isometry V (Definition 2.5) and port Grams $\{A_k\}$. Define

$$M_S \coloneqq V^\dagger \Big[\prod_{k \in S} (\mathbb{1} - \Pi_k^0) \prod_{k \notin S} \Pi_k^0 \Big] V, \qquad S \subseteq [K]. \tag{4.5}$$

Then:

(i) *$\{M_S\}$ is a POVM.* $M_S \succeq 0$ for every S and $\sum_{S \subseteq [K]} M_S = \mathbb{1}$. Moreover $\Pi_{\mathrm{vac}}(J) \coloneqq V^\dagger \Pi_J^0 V = \sum_{S \subseteq [K]\setminus J} M_S$ for every $J \subseteq [K]$.

(ii) *Closed form.* $\Pi_{\mathrm{vac}}(J) = \mathfrak{F}(\mathbb{1}_{\mathbb{I}} - A_J)$ and

$$M_S = \sum_{T \subseteq S} (-1)^{|T|}\, \mathfrak{F}\big(\mathbb{1}_{\mathbb{I}} - A_{([K]\setminus S)\cup T}\big). \tag{4.6}$$

(iii) *Photon-number block structure.* Every M_S commutes with the total photon number of the declared inputs, so $M_S = \bigoplus_{n \geq 0} M_S^{(=n)}$ with $M_S^{(=n)}$ acting on $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$, and

$$M_S^{(=n)} = \sum_{T \subseteq S} (-1)^{|T|} \Big(\big(\mathbb{1} - A_{([K]\setminus S)\cup T}\big)^{\otimes n} \Big) \Big|_{\mathrm{Sym}^n}. \tag{4.7}$$

In particular $M_S^{((=0))} = \delta_{S,\varnothing}$: no photon, no click.

(iv) *The coarse elements.* With $A_\Sigma = A_{[K]}$,

$$M_{\mathrm{nc}} \coloneqq M_\varnothing = \mathfrak{F}(\mathbb{1} - A_\Sigma), \tag{4.8}$$

$$M_{\mathrm{sc}_k} \coloneqq M_{\{k\}} = \mathfrak{F}\big(\mathbb{1} - A_{[K]\setminus\{k\}}\big) - \mathfrak{F}(\mathbb{1} - A_\Sigma), \tag{4.9}$$

$$M_{\mathrm{mc}} \coloneqq \sum_{S \in \mathsf{MC}} M_S = \mathbb{1} + (K-1)\,\mathfrak{F}(\mathbb{1} - A_\Sigma) - \sum_{k=1}^{K} \mathfrak{F}\big(\mathbb{1} - A_{[K]\setminus\{k\}}\big). \tag{4.10}$$

5 Representation independence and coarse-graining

Theorem 5.1 (Representation independence of the click-pattern POVM)**.** Let $\mathfrak{R}, \mathfrak{R}'$ be receiver instruments on the same declared input space $\mathbb{C}^{\mathbb{I}}$ with the same number of ports K.

(i) *Determination, in both directions.* $\mathfrak{R}$ and $\mathfrak{R}'$ induce the same click-pattern POVM if and only if $A_k = A'_k$ for every k. In particular the POVM depends on the instrument matrix only through the port Grams.

(ii) *Realizability.* Every admissible family $\{A_k\}_{k=1}^{K}$ on $\mathbb{C}^{\mathbb{I}}$ (Definition 2.2) is the port-Gram family of some receiver instrument, in which port k collects at most $\operatorname{rank} A_k \le |\mathbb{I}|$ modes.

(iii) *Port-local unitaries are invisible.* If $G' = \big(\bigoplus_k V_k \oplus V_{\mathrm{rest}}\big)G$ with each V_k unitary on $\mathbb{C}^{\mathbb{P}_k}$ and V_{rest} unitary on the remaining modes, then $A'_k = A_k$ for every k.

(iv) *One mode per port.* There is a receiver instrument with $|\mathbb{P}_k| = 1$ for every k inducing the same POVM as $\mathfrak{R}$ if and only if $\operatorname{rank} A_k \le 1$ for every k. In that case, writing $A_k = \lambda_k u_k u_k^\dagger$ with $\lambda_k \ge 0$ and u_k a unit vector, the single row of port k is $g_k = e^{i\varphi_k}\sqrt{\lambda_k}\, u_k^\dagger$, unique up to the phase φ_k.

Corollary 5.2 (Unobservability of per-port phases)**.** (i) Multiplying the output row of a port by a phase, or more generally applying any unitary *inside* a single port's mode set, changes no click statistic (Theorem 5.1 (iii)).

(ii) *The hypothesis matters*: the unitary must act inside one port's modes and after all interference has taken place. A phase that is global on an element in the *middle* of the receiver can become relative once that element's output paths are recombined with other paths, and it can then change the Grams.

Corollary 5.3 (Extinction criterion for rank one at two input modes)**.** Suppose $|\mathbb{I}| = 2$. Then $\operatorname{rank} A_k \le 1$ holds if and only if either $A_k = 0$ or there is a unit vector $v \in \mathbb{C}^2$ with $v^\dagger A_k v = 0$, that is, a polarization the port extinguishes completely. For $|\mathbb{I}| > 2$ the two conditions are *not* equivalent: the existence of one null direction of a positive semidefinite A_k says only $\operatorname{rank} A_k \le |\mathbb{I}| - 1$.

Definition 5.5 (Declared coarse-graining and the emitted POVM)**.** A *declared coarse-graining* is a surjection $\varpi : 2^{[K]} \to \mathcal{V}$ onto the declared emitted alphabet, supplied as data by the protocol declaration, $[K]$ indexing the detector ports of a receiver instrument (Definition 2.1). With M_S the click-pattern elements of Theorem 4.5 ((4.5)), the *emitted POVM* is

$$F_v \coloneqq \sum_{S \in \varpi^{-1}(v)} M_S, \qquad v \in \mathcal{V}. \tag{5.2}$$

Theorem 5.6 (Coarse-graining of the click-pattern POVM). Let ϖ be a declared coarse-graining.

(i) $\{F_v\}_{v\in\mathcal{V}}$ is a POVM on $\mathcal{F}(\mathbb{C}^{\mathbb{I}})$, and every F_v is block diagonal in the photon number with $F_v^{(=n)} = \sum_{S\in\varpi^{-1}(v)} M_S^{(=n)}$.

(ii) *Bounds add.* If $\mathcal{O} \subseteq 2^{[K]}$ and $M_S^{(>N)} \succeq p_S \mathbb{1}^{(>N)}$ for every $S \in \mathcal{O}$ with $p_S \geq 0$, then $\sum_{S\in\mathcal{O}} M_S^{(>N)} \succeq \big(\sum_{S\in\mathcal{O}} p_S\big)\mathbb{1}^{(>N)}$. In particular, if $\varpi^{-1}(v) \supseteq \mathcal{O}$ then $F_v^{(>N)} \succeq \big(\sum_{S\in\mathcal{O}} p_S\big)\mathbb{1}^{(>N)}$.

(iii) *Coarsening is lossy in one direction only.* A bound on a coarse outcome does not descend to its fibres: $F_v \succeq p\,\mathbb{1}$ implies nothing about the individual M_S with $\varpi(S) = v$.

6 The analytic high-photon bound

Definition 6.1 (Admissible bound data and the bound function)**.** A tuple $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ with $K \geq 1$ entries is *admissible bound data* when

$$0 \leq \gamma \leq \beta_k \leq 1, \qquad 0 \leq c \leq d_k \leq 1 \qquad (k = 1, \ldots, K), \tag{6.1}$$

and the associated *bound function* is

$$\beta(n) \coloneqq 1 + (K-1)\,\gamma\, c^n - \sum_{k=1}^{K} \beta_k\, d_k^n, \qquad n \geq 0, \tag{6.2}$$

with the convention $0^0 = 1$. Admissibility (6.1) is exact. The one part of it that this contract decides on computed data is $0 \leq c$; the other parts are not separately tested (Definition 10.4 (iv)). In floating point that part is decided by the widened test $c \geq -\varepsilon_2$ of Definition 10.4 (iv), which decides it on *every* input route and is not displaced by the route-(ii) admission of Definition 10.4 (v); the computed c enters (6.2) unclamped.

Theorem 6.2 (Lower bound on every multiclick block)**.** Let $B, B_1, \ldots, B_K$ be Hermitian operators on a finite-dimensional space with

$$0 \preceq B_k \preceq B \preceq \mathbb{1} \qquad (k = 1, \ldots, K), \tag{6.3}$$

let $\gamma, \beta_1, \dots, \beta_K \in (0,1]$ with $\gamma \le \beta_k$ for every k, let $\Pi_{\mathrm{vac}}([K])$ and $\Pi_{\mathrm{vac}}([K]\setminus\{k\})$ $(k = 1, \dots, K)$ be operators on the Fock space of that space of the form

$$\Pi_{\mathrm{vac}}([K]) = \gamma\, \mathfrak{F}(\mathbb{1} - B), \qquad \Pi_{\mathrm{vac}}([K] \setminus \{k\}) = \beta_k\, \mathfrak{F}\big(\mathbb{1} - B_k\big), \tag{6.4}$$

and put $M_{\mathrm{mc}} \coloneqq \mathbb{1} + (K-1)\Pi_{\mathrm{vac}}([K]) - \sum_k \Pi_{\mathrm{vac}}([K] \setminus \{k\})$. Put

$$c \coloneqq 1 - \lambda_{\max}(B), \qquad d_k \coloneqq 1 - \lambda_{\min}\big(B_k\big). \tag{6.5}$$

Then $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ is admissible bound data, and for every $n \ge 0$

$$\lambda_{\min}\big(M_{\mathrm{mc}}^{(=n)}\big) \;\ge\; \beta(n), \qquad \text{equivalently} \qquad M_{\mathrm{mc}}^{(=n)} \;\succeq\; \beta(n)\, \mathbb{1}^{(=n)}. \tag{6.6}$$

Moreover $\beta(n) \le 1$ for every n.

Lemma 6.5 (Collapse of the tail infimum to the first photon number)**.** Let $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ be admissible bound data satisfying

$$\sum_{k=1}^{K} \beta_k \;\ge\; 1 + (K-1)\gamma \qquad (\text{equivalently, } \beta(0) \le 0). \tag{6.8}$$

Then for every $n \ge 1$,

$$\beta(n) \ge 0 \implies \beta(n+1) \;\ge\; \beta(n). \tag{6.9}$$

Consequently, for every cutoff $N \ge 0$,

$$\max\Big\{0,\ \inf_{n>N} \beta(n)\Big\} \;=\; \max\big\{0,\ \beta(N+1)\big\}, \tag{6.10}$$

and if $\beta(N+1) \ge 0$ then β is non-decreasing on $\{N+1, N+2, \dots\}$ and $\inf_{n>N} \beta(n) = \beta(N+1)$. Hypothesis (6.8) holds automatically when $\gamma = \beta_k = 1$, where it reads $K \ge K$.

Lemma 6.8 (Window-and-tail bound on the tail infimum)**.** Let $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ be admissible bound data and $N \ge 0$. Define, for an integer $m > N$,

$$\underline{\beta}_{N,m} \coloneqq \min\big\{\, W_{N,m},\, T_m \,\big\}, \qquad W_{N,m} \coloneqq \min_{N<n\le m} \beta(n), \qquad T_m \coloneqq 1 - \sum_{k=1}^{K} \beta_k d_k^m. \tag{6.18}$$

The window $\{n : N < n \le m\}$ is non-empty for every $m > N$; at $m = N+1$ it is the single index $N+1$. Then:

(i) $\underline{\beta}_{N,m} \le \inf_{n>N} \beta(n)$ for every $m > N$: the quantity is a valid uniform bound;

(ii) $\underline{\beta}_{N,m}$ is non-decreasing in m, and $\lim_{m\to\infty}\underline{\beta}_{N,m} = \inf_{n>N}\beta(n)$ whenever $c<1$; moreover the *residual gap*

$$g_{N,m} \coloneqq \max\{0,\ W_{N,m} - T_m\} \tag{6.19}$$

is a finite non-negative real for every $m > N$ — at $m = N+1$ it equals $(K-1)\gamma c^{N+1}$ — and satisfies $\inf_{n>N}\beta(n) - \underline{\beta}_{N,m} \le g_{N,m}$, so the evaluation is convergent and $g_{N,m}$ bounds what widening the window could still recover;

(iii) if $c = 0$ then $\beta(n) = 1 - \sum_k \beta_k d_k^n$ for $n \ge 1$ is non-decreasing, $\underline{\beta}_{N,N+1} = \beta(N+1)$ is exact, and $g_{N,N+1} = 0$;

(iv) the two lemmas are consistent: under the hypothesis of Lemma 6.5 with $\beta(N+1) \ge 0$ *and* $c<1$, (i) and (ii) give $\underline{\beta}_{N,m} \le \beta(N+1)$ for every m with equality in the limit $m\to\infty$, so the first branch of Definition 8.3 is the exact value the window-and-tail evaluation converges to, and using it loses nothing. The hypothesis $c<1$ is needed: at $c=1$ admissibility forces $d_k = 1$ for every k, β is constant, and under Lemma 6.5's hypothesis with $\beta(N+1)\ge 0$ that constant is 0 while $T_m = 1 - \sum_k \beta_k = -(K-1)\gamma$ for every m, so the limit clause fails whenever $\gamma > 0$ and $K \ge 2$.

7 Exact block eigenvalues

Definition 7.1 (The exact block bound function)**.** Under the hypotheses of Theorem 6.2, the *exact block bound* is

$$\beta^*(n) \coloneqq \lambda_{\min}\big(M_{\mathrm{mc}}^{(=n)}\big) = \lambda_{\min}\Big(\mathbb{1}^{(=n)} + (K-1)\gamma\,\mathfrak{F}(\mathbb{1}-B)^{(n)} - \sum_{k=1}^{K}\beta_k\,\mathfrak{F}\big(\mathbb{1}-B_k\big)^{(n)}\Big), \tag{7.1}$$

for $n \ge 0$, the second equality being (6.4) restricted to the n-photon block. The operator inside is a Hermitian matrix of dimension $\binom{D+n-1}{n}$, where D is the dimension of the space B acts on: the channel input space of the setting family (Definition 8.1), which is $|\mathbb{I}_C|$ at both instances this module constructs. No new object is introduced: each $\mathfrak{F}(\cdot)^{(n)} = \big((\cdot)^{\otimes n}\big)\big|_{\mathrm{Sym}^n}$ is the construction (4.7) already uses for the blocks the artifact emits at $n \le N$, in the occupation basis and order that Definition 1.2 and the interface conventions of this module fix, evaluated at an index above the cutoff instead of below it.

Lemma 7.2 (Range of the exact block bound)**.** Let the hypotheses of Theorem 6.2 hold. Then, for every $n \ge 0$,

$$\beta(n) \le \beta^*(n) \le 1. \tag{7.2}$$

If in addition

$$M_{\mathrm{mc}} \succeq 0, \tag{7.3}$$

then $\beta^*(n) \geq 0$ as well, so that $\beta^*(n) \in [0,1]$.

Lemma 7.4 (Exact window with an analytic remainder)**.** Let the hypotheses of Theorem 6.2 hold together with (7.3), let $N \geq 0$, and let $m^* > N$ be an integer. Let $\ell(n)$, $N < n \leq m^*$, be reals with

$$\ell(n) \leq \beta^*(n) \qquad \text{for } N < n \leq m^*, \tag{7.5}$$

and let τ be a real with

$$\tau \leq 0 \qquad \text{or} \qquad \tau \leq \inf_{n > m^*} \beta^*(n). \tag{7.6}$$

Then

$$P := \max\Big\{0,\ \min\Big\{\ \min_{N<n\leq m^*} \ell(n),\ \ \tau\ \Big\}\Big\} \tag{7.7}$$

satisfies $0 \leq P \leq 1$ and

$$M_{\mathrm{mc}}^{(>N)} \succeq P\, \mathbb{1}^{(>N)}. \tag{7.8}$$

Three choices are admissible:

(i) $\tau = T_{m^*}$ of (6.18), formed from the bound data $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ that Theorem 6.2 forms by (6.5), satisfies (7.6) with no further hypothesis;

(ii) if the bound data satisfy (6.8), then $\tau = \beta(m^*+1)$ satisfies (7.6);

(iii) $\ell(n) = \beta(n)$ satisfies (7.5), and the pointwise maximum of finitely many admissible choices of ℓ is admissible.

8 Setting families and the emitted weight

Definition 8.1 (Setting family)**.** Let $D \geq 1$ and write $\mathcal{F}_D := \mathcal{F}(\mathbb{C}^D)$ for the Fock space of Definition 1.2 over a D-dimensional *channel input space*, with photon-number projectors Π_n onto $\mathrm{Sym}^n(\mathbb{C}^D)$. A *setting family* of $K \geq 1$ ports on $\mathcal{F}_D$ is a finite non-empty set $\mathcal{T}_{\mathrm{set}}$ of *settings* together with, for every $t \in \mathcal{T}_{\mathrm{set}}$:

(a) a POVM $\{M_{S|t}\}_{S\subseteq[K]}$ on $\mathcal{F}_D$ every element of which commutes with the photon number, so that $M_{S|t} = \bigoplus_{n\geq 0} M_{S|t}^{(=n)}$ with $M_{S|t}^{(=n)}$ acting on $\mathrm{Sym}^n(\mathbb{C}^D)$. Its *joint no-click elements* and its *multiclick element* are

$$\Pi_{\mathrm{vac}}(J)^{(t)} := \sum_{S\subseteq[K]\setminus J} M_{S|t} \quad (J \subseteq [K]), \qquad M_{\mathrm{mc}|t} := \sum_{S\in\mathsf{MC}} M_{S|t}, \tag{8.1}$$

with MC the multiclick set of Definition 4.4;

(b) Hermitian operators $B^{(t)}, B_1^{(t)}, \dots, B_K^{(t)}$ on $\mathbb{C}^D$ and scalars $\gamma^{(t)}, \beta_1^{(t)}, \dots, \beta_K^{(t)} \in (0,1]$ satisfying the hypotheses of Theorem 6.2, namely (6.3) and $\gamma^{(t)} \le \beta_k^{(t)}$ for every k, such that the joint no-click elements have the prefactor form (6.4),

$$\Pi_{\mathrm{vac}}([K])^{(t)} = \gamma^{(t)}\,\mathfrak{F}\big(\mathbb{1}-B^{(t)}\big), \qquad \Pi_{\mathrm{vac}}([K]\backslash\{k\})^{(t)} = \beta_k^{(t)}\,\mathfrak{F}\big(\mathbb{1}-B_k^{(t)}\big); \tag{8.2}$$

(c) a flag $\chi^{(t)} \in \{\text{exact}, \text{general}\}$, the setting's *exact flag*, which may be exact only if $\gamma^{(t)} = \beta_k^{(t)} = 1$ for every k holds *exactly, by the construction that produced the setting*, read off the declaration with no arithmetic performed, so that at an exact setting (6.8) reads $K \ge K$ and holds with equality by construction.

The *bound data of the setting* are $\big(\gamma^{(t)}, \{\beta_k^{(t)}\}, c^{(t)}, \{d_k^{(t)}\}\big)$ with $c^{(t)} \coloneqq 1 - \lambda_{\max}(B^{(t)})$ and $d_k^{(t)} \coloneqq 1 - \lambda_{\min}(B_k^{(t)})$ as in (6.5); they are admissible bound data (Definition 6.1) by Theorem 6.2, and $\beta^{(t)}$ denotes their bound function (6.2).

Two consequences are used throughout. First, the multiclick element has the inclusion–exclusion form

$$M_{\mathrm{mc}|t} = \mathbb{1} + (K-1)\,\Pi_{\mathrm{vac}}([K])^{(t)} - \sum_{k=1}^{K} \Pi_{\mathrm{vac}}([K]\setminus\{k\})^{(t)}, \tag{8.3}$$

because completeness gives $M_{\mathrm{mc}|t} = \mathbb{1} - M_{\varnothing|t} - \sum_k M_{\{k\}|t}$ while (8.1) gives $M_{\varnothing|t} = \Pi_{\mathrm{vac}}([K])^{(t)}$ and $M_{\{k\}|t} = \Pi_{\mathrm{vac}}([K]\setminus\{k\})^{(t)} - \Pi_{\mathrm{vac}}([K])^{(t)}$; so Theorem 6.2 applies to every setting, with $\lambda_{\min}(M_{\mathrm{mc}|t}^{(=n)}) \ge \beta^{(t)}(n)$ for every $n \ge 0$. Second, $M_{\mathrm{mc}|t}$ is a sum of elements of a POVM, so hypothesis (7.3) of Lemma 7.2, $M_{\mathrm{mc}|t} \succeq 0$, holds at every setting.

Corollary 8.2 (The unconditioned receiver is a setting family)**.** Let $\mathfrak{R}$ be a receiver instrument (Definition 2.1) with $\mathbb{I}_P = \varnothing$, so that $\mathbb{I} = \mathbb{I}_C$, and let $\{A_k\}$ be its port Grams and $\{M_S\}$ its click-pattern POVM (Theorem 4.5). Then $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$ with

$$M_{S|\mathrm{off}} \coloneqq M_S, \qquad B^{(\mathrm{off})} \coloneqq A_\Sigma, \qquad B_k^{(\mathrm{off})} \coloneqq A_{[K]\setminus\{k\}}, \qquad \gamma^{(\mathrm{off})} = \beta_k^{(\mathrm{off})} \coloneqq 1, \qquad \chi^{(\mathrm{off})} \coloneqq \text{exact}, \tag{8.4}$$

is a setting family (Definition 8.1) on $\mathcal{F}(\mathbb{C}^{\mathbb{I}})$, with $D = |\mathbb{I}|$. Its bound data are $c = 1 - \lambda_{\max}(A_\Sigma)$ and $d_k = 1 - \lambda_{\min}(A_{[K]\setminus\{k\}})$, and (6.8) holds with equality.

Definition 8.3 (The uniform high-photon multiclick weight)**.** Fix a cutoff N and a setting family (Definition 8.1); for each setting $t \in \mathcal{T}_{\mathrm{set}}$ let $\beta^{(t)}$

be the bound function (6.2) of the bound data of the setting. At every setting t whose declaration carries *no* exact-window depth (Definition 8.7), the *per-setting weight* is

$$p_{\mathrm{mc}}^{(t)} := \begin{cases} \max\{0,\ \beta^{(t)}(N+1)\} & \text{if (X) holds at } t, \\ \max\{0,\ \underline{\beta}_{N,m_t}^{(t)}\} & \text{otherwise, for the declared window } m_t > N, \end{cases} \tag{8.5}$$

and the *uniform weight* is $p_{\mathrm{mc}} := \min_{t\in\mathcal{T}_{\mathrm{set}}} p_{\mathrm{mc}}^{(t)}$, where the branch-1 *predicate (X)* is

(X) ***exact by construction, decidable from the declaration:*** the setting's exact flag is $\chi^{(t)} = \text{exact}$ (Definition 8.1 (c)). Then $\gamma^{(t)} = \beta_k^{(t)} = 1$ exactly, so (6.8) reads $K \geq K$ and holds with *equality*, no arithmetic having been performed.

Where the exact window replaces this. At a setting whose declaration carries an exact-window depth m_t^*, (8.5) does not apply and $p_{\mathrm{mc}}^{(t)}$ is (8.7) instead (Definition 8.7); $p_{\mathrm{mc}} = \min_{t\in\mathcal{T}_{\mathrm{set}}} p_{\mathrm{mc}}^{(t)}$ is unchanged, and the two kinds of setting may be mixed in one declaration. A declaration carrying no depth at any setting is governed by (8.5) alone: no β^* is formed, no eigenvalue of any block above the cutoff is computed, and Assumption 11.11 is not invoked.

Definition 8.7 (Exact-window weight of a setting with a declared depth)**.** Let $t \in \mathcal{T}_{\mathrm{set}}$ be a setting whose declaration carries an *exact-window depth* m_t^*: an integer with $m_t^* > N$ and, where predicate (X) fails at t, also $m_t^* \leq m_t$. Write $\widehat{\beta}^{*(t)}(n)$ for the value an implementation computes for $\beta^{*(t)}(n)$ of (7.1) at that setting, let $\varepsilon_{\mathrm{eig}}$ be the declared *eigenvalue slack*, and put

$$\ell^{(t)}(n) \ :=\ \max\left\{\ \beta^{(t)}(n),\ \ \widehat{\beta}^{*(t)}(n) - \varepsilon_{\mathrm{eig}}\ \right\}, \qquad N < n \leq m_t^*. \tag{8.6}$$

Then

$$p_{\mathrm{mc}}^{(t)} := \begin{cases} \max\left\{0,\ \min\left\{\ \min\limits_{N<n\leq m_t^*} \ell^{(t)}(n),\ \ \beta^{(t)}\big(m_t^*+1\big)\ \right\}\right\} & \text{if (X) holds at } t, \\ \max\left\{0,\ \min\left\{\ \min\limits_{N<n\leq m_t^*} \ell^{(t)}(n),\ \ \min\limits_{m_t^*<n\leq m_t} \beta^{(t)}(n),\ \ T_{m_t}^{(t)}\ \right\}\right\} & \text{otherwise,} \end{cases} \tag{8.7}$$

the middle minimum in the second case running over the empty index set — and so equal to $+\infty$ and inactive — when $m_t^* = m_t$.

Per setting. The depth is optional and per setting, and each per-setting weight $p_{\mathrm{mc}}^{(t)}$ is its own setting's: a depth changes the weight of the setting that declares it and of no other. A declaration that carries a depth at some settings and not at others therefore obtains the exact-window value as the weight of each setting with one, while the uniform weight $p_{\mathrm{mc}} = \min_{t\in\mathcal{T}_{\mathrm{set}}} p_{\mathrm{mc}}^{(t)}$ is bounded by the settings without one.

Proposition 8.9 (Dominance of the exact-window weight over the analytic weight)**.** At every setting carrying a declared exact-window depth, the value (8.7) is at least the value (8.5) gives at that setting on the same bound data.

Definition 8.12 (The weight evaluated on enclosed bound data)**.** Fix a cutoff N, a setting family (Definition 8.1) and a setting $t \in \mathcal{T}_{\mathrm{set}}$, and write $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ for the bound data of the setting, the index t omitted.

(i) Reals $\big(\check{\gamma}, \{\check{\beta}_k\}, \check{c}, \{\check{d}_k\}\big)$ are *enclosed bound data* of t when

$$0 \le \check{\gamma} \le \gamma, \qquad \beta_k \le \check{\beta}_k \le 1, \qquad 0 \le \check{c} \le c, \qquad d_k \le \check{d}_k \le 1 \qquad (k = 1, \dots, K). \tag{8.8}$$

$\check{\beta}$ denotes the bound function (6.2) of this tuple, and $\check{T}_m$ and $\underline{\check{\beta}}_{N,m}$ the quantities T_m and $\underline{\beta}_{N,m}$ of (6.18) formed from it.

(ii) Where t carries an exact-window depth m_t^* (Definition 8.7), reals $w(n)$, $N < n \le m_t^*$, are *rigorous window bounds* of t when $w(n) \le \beta^{*(t)}(n)$ for each such n, $\beta^{*(t)}$ being the exact block bound (7.1) at t.

(iii) The *enclosed weight* $\check{p}_{\mathrm{mc}}^{(t)}$ of t formed from them is (8.5) where t carries no exact-window depth and (8.7) where it carries one, its branch chosen by predicate (X) at t as there, each read with $\check{\beta}$, $\underline{\check{\beta}}_{N,m_t}$ and $\check{T}_{m_t}$ in place of $\beta^{(t)}$, $\underline{\beta}^{(t)}_{N,m_t}$ and $T^{(t)}_{m_t}$, and with

$$\check{\ell}(n) \;:=\; \max\big\{\check{\beta}(n),\; w(n)\big\}, \qquad N < n \le m_t^*, \tag{8.9}$$

in place of $\ell^{(t)}(n)$ of (8.6). No computed eigenvalue of a block and no eigenvalue slack enters it.

Theorem 8.13 (The weight on enclosed bound data bounds the multiclick element)**.** Let a setting family (Definition 8.1) be given, let $N \ge 0$, and let $t \in \mathcal{T}_{\mathrm{set}}$, with operators $B = B^{(t)}$ and $B_k = B_k^{(t)}$, bound data of the setting $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$, bound function $\beta = \beta^{(t)}$ and T_m of (6.18) formed from those data. Write $\mathrm{clip}(x) := \min\{1, \max\{0, x\}\}$ for real x.

(i) *Construction.* Let u, $\ell_1, \dots, \ell_K$, g and $b_1, \dots, b_K$ be reals with $u \ge \lambda_{\max}(B)$, $\ell_k \le \lambda_{\min}(B_k)$, $g \le \gamma$ and $b_k \ge \beta_k$. Then

$$\big(\,\mathrm{clip}(g),\; \{\mathrm{clip}(b_k)\},\; \mathrm{clip}(1-u),\; \{\mathrm{clip}(1-\ell_k)\}\big) \tag{8.10}$$

are enclosed bound data of t. Where t carries an exact-window depth m_t^* and reals λ_n satisfy $\lambda_n \le \lambda_{\min}\big(M_{\mathrm{mc}|t}^{(=n)}\big)$ for $N < n \le m_t^*$, the reals $\max\{0, \lambda_n\}$ are rigorous window bounds of t.

(ii) *Comparison.* Enclosed bound data $\big(\check\gamma, \{\check\beta_k\}, \check c, \{\check d_k\}\big)$ of t are admissible bound data (Definition 6.1), and their bound function $\check\beta$ and $\check T_m$ satisfy

$$\check\beta(n) \;\le\; \beta(n) \quad (n \ge 0), \qquad \check T_m \;\le\; T_m \quad (m \ge 0). \tag{8.11}$$

(iii) *The bound.* For enclosed bound data of t and, where t carries an exact-window depth, rigorous window bounds of t, the enclosed weight $\check p_{\mathrm{mc}}^{(t)}$ formed from them (Definition 8.12) satisfies $0 \le \check p_{\mathrm{mc}}^{(t)} \le 1$ and

$$M_{\mathrm{mc}|t}^{(>N)} \;\succeq\; \check p_{\mathrm{mc}}^{(t)}\, \mathbb{1}^{(>N)}, \tag{8.12}$$

and every real p with $0 \le p \le \check p_{\mathrm{mc}}^{(t)}$ satisfies $0 \le p \le 1$ and $M_{\mathrm{mc}|t}^{(>N)} \succeq p\, \mathbb{1}^{(>N)}$.

Theorem 8.14 (Multiclick bounds carried by the emitted weights). Let a setting family (Definition 8.1) be given. For every setting $t \in \mathcal{T}_{\mathrm{set}}$ let $p_{\mathrm{mc}}^{(t)}$ be as in Definition 8.3 if t carries no exact-window depth and as in Definition 8.7 if it carries one, or, at any setting, let it be the enclosed weight $\check p_{\mathrm{mc}}^{(t)}$ of Definition 8.12 formed from enclosed bound data of t and, where t carries a depth, from rigorous window bounds of t, or a real p with $0 \le p \le \check p_{\mathrm{mc}}^{(t)}$; and let $p_{\mathrm{mc}} = \min_{t \in \mathcal{T}_{\mathrm{set}}} p_{\mathrm{mc}}^{(t)}$ as in Definition 8.3. Then

$$M_{\mathrm{mc}|t}^{(>N)} \;\succeq\; p_{\mathrm{mc}}^{(t)}\, \mathbb{1}^{(>N)} \qquad \text{for every } t \in \mathcal{T}_{\mathrm{set}}, \tag{8.13}$$

where $M_{\mathrm{mc}|t}^{(>N)}$ is the restriction of the multiclick element $M_{\mathrm{mc}|t}$ of (8.1) at setting t to the photon-number blocks $n > N$; and consequently, with the one weight p_{mc} at every setting,

$$M_{\mathrm{mc}|t}^{(>N)} \;\succeq\; p_{\mathrm{mc}}\, \mathbb{1}^{(>N)} \qquad \text{for every } t \in \mathcal{T}_{\mathrm{set}}. \tag{8.14}$$

At a setting governed by Definition 8.3 or by Definition 8.12 both inequalities hold at that setting with no hypothesis beyond this contract's ledger; at a setting governed by Definition 8.7 they hold there *under Assumption 11.11*, which is what makes the second entry of (8.6) a lower bound. At a one-setting family, $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$, one has $p_{\mathrm{mc}} = p_{\mathrm{mc}}^{(\mathrm{off})}$ and the two inequalities are one. Moreover $0 \le p_{\mathrm{mc}}^{(t)} \le 1$ at every setting, hence $0 \le p_{\mathrm{mc}} \le 1$, and each $p_{\mathrm{mc}}^{(t)}$ of Definition 8.3 or 8.7 is computable by finitely many arithmetic operations on $\{\lambda_{\max}(B^{(t)}), \lambda_{\min}(B_k^{(t)}), \gamma^{(t)}, \beta_k^{(t)}\}$ and the declared N, m_t and m_t^* — where $B^{(t)}, B_k^{(t)}$ are the operators of the setting (Definition 8.1 (b)) — together with, where the setting carries a declared depth, the declared $\varepsilon_{\mathrm{eig}}$ and the $m_t^* - N$ extreme eigenvalues $\widehat{\beta}^{*(t)}(n)$ of the Hermitian matrices (7.1), each of dimension $\binom{D+n-1}{n}$; each enclosed weight by finitely many on the enclosed bound data, the rigorous window bounds and the declared N, m_t and m_t^*; and p_{mc} by finitely many comparisons among them.

Corollary 8.15 (Emitted multiclick weight of a probe-free receiver)**.** Let $\mathfrak{R}$ be a receiver instrument with $\mathbb{I}_P = \varnothing$ (Definition 2.1), let N be the declared cutoff, and let p_{mc} be computed by Definition 8.3, or by Definition 8.7 where an exact-window depth is declared, from the one-setting family of Corollary 8.2, or let it be a real with $0 \le p_{\mathrm{mc}} \le \check{p}_{\mathrm{mc}}$, where $\check{p}_{\mathrm{mc}}$ is an enclosed weight of that family's one setting (Definition 8.12), $\check{p}_{\mathrm{mc}}$ itself included. Then

$$M_{\mathrm{mc}}^{(>N)} \;\succeq\; p_{\mathrm{mc}}\, \mathbb{1}^{(>N)}, \qquad 0 \le p_{\mathrm{mc}} \le 1, \tag{8.15}$$

with no hypothesis beyond the ledger where p_{mc} is given by Definition 8.3 or 8.12 and under Assumption 11.11 where it is given by Definition 8.7; and the strengthening slot $\{(\mathsf{MC}, p_{\mathrm{mc}})\}$ satisfies the guarantee of Definition 8.16.

Definition 8.16 (The strengthening slot)**.** The *strengthening slot* is the finite list of pairs in which the artifact carries the high-photon evidence,

$$\Big\{\big(\mathcal{O}_i,\ (p_i^{(t)})_{t\in\mathcal{T}_{\mathrm{set}}}\big)\Big\}_{i=1}^{r}, \qquad \mathcal{O}_i \subseteq 2^{[K]} \text{ pairwise disjoint}, \quad p_i^{(t)} \ge 0, \tag{8.16}$$

one weight per entry and per setting $t \in \mathcal{T}_{\mathrm{set}}$ of the setting family (Definition 8.1), with the guarantee $\sum_{S\in\mathcal{O}_i} M_{S|t}^{(>N)} \succeq p_i^{(t)}\, \mathbb{1}^{(>N)}$ for every setting $t \in \mathcal{T}_{\mathrm{set}}$ and every i. The *uniform weight of an entry* i is $p_i := \min_{t\in\mathcal{T}_{\mathrm{set}}} p_i^{(t)}$; since $0 \le p_i \le p_i^{(t)}$, it carries the guarantee $\sum_{S\in\mathcal{O}_i} M_{S|t}^{(>N)} \succeq p_i\, \mathbb{1}^{(>N)}$ for every setting $t \in \mathcal{T}_{\mathrm{set}}$ and every i, one weight at every setting. At a one-setting family, $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$, an entry has the single weight $p_i^{(\mathrm{off})} = p_i$ and is written $(\mathcal{O}_i, p_i)$. The artifact carries $r = 1$ with $(\mathcal{O}_1, (p_1^{(t)})_{t\in\mathcal{T}_{\mathrm{set}}}) = (\mathsf{MC}, (p_{\mathrm{mc}}^{(t)})_{t\in\mathcal{T}_{\mathrm{set}}})$, the per-setting weights of Theorem 8.14, each given at its setting by Definition 8.3 or 8.7 or as a non-negative real at most the enclosed weight of Definition 8.12, whose uniform weight is p_{mc}; at a one-setting family that entry is $(\mathsf{MC}, p_{\mathrm{mc}})$.

Proposition 8.17 (Range of the strengthening-slot weights). For any list (8.16) satisfying the stated guarantee with pairwise disjoint $\mathcal{O}_i$, one has at every setting $t \in \mathcal{T}_{\mathrm{set}}$ that $p_i^{(t)} \ge 0$ for every i and $\sum_i p_i^{(t)} \le 1$; hence the uniform weights satisfy $p_i \ge 0$ for every i and $\sum_i p_i \le 1$. In particular the list this module emits — by Corollary 8.15 and, for a setting family a submodule constructs, by that submodule's instance of Theorem 8.14 — satisfies $p_{\mathrm{mc}}^{(t)} \in [0,1]$ at every setting, so the failure conditions "$p_y < 0$" and "$\sum_y p_y > 1$" of a squashing request cannot be triggered by this module's output; and what the further failure condition of a squashing request, "a bound does not hold at every declared setting it is stated for and every photon number above the cutoff", requires to hold — each $p_{\mathrm{mc}}^{(t)}$ at its own setting t and p_{mc} at every setting, for every photon number above the cutoff — is Theorem 8.14.

9 The reference, enclosed and exact

Definition 9.1 (Complex enclosures, and enclosures of square roots and angles)**.** Write $\mathbb{Q}(i) = \{x + iy : x, y \in \mathbb{Q}\}$ and, for $z \in \mathbb{Q}(i)$, $|z|_1 := |\mathrm{Re}\, z| + |\mathrm{Im}\, z|$, a rational with $|z| \le |z|_1$. A *complex enclosure* of a complex number x is a pair (x_c, r) of $x_c \in \mathbb{Q}(i)$, its *centre*, and a rational $r \ge 0$, its *radius*, with $|x - x_c| \le r$. It is *exact* when $r = 0$, and then $x = x_c$. A complex enclosure of a complex matrix X is a pair (X_c, r) of a matrix X_c of the same shape with entries in $\mathbb{Q}(i)$ and a matrix r of non-negative rationals such that $\big((X_c)_{il}, r_{il}\big)$ is a complex enclosure of X_{il} for all i, l.

A *square-root enclosure* of a rational $p \ge 0$ is a complex enclosure of $\sqrt{p}$ with a rational centre, exact where p is the square of a rational. An *angle enclosure* of a real number θ that is given either as a rational, such as a declared value (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)), or as $c\pi$ with c a rational, is a pair of complex enclosures with rational centres of $\cos\theta$ and of $\sin\theta$, both exact where $\theta = c\pi$ with $2c$ an integer.

Theorem 9.5 (Rigorous enclosures of the photon-number blocks)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}_{k=1}^{K}$ on $\mathbb{C}^{\mathbb{I}}$ (Definitions 2.1 and 2.2), $\{M_S\}_{S\subseteq[K]}$ its click-pattern POVM (Theorem 4.5), ϖ a declared coarse-graining onto $\mathcal{V}$ and $\{F_v\}_{v\in\mathcal{V}}$ the emitted POVM (Definition 5.5). Fix $N \ge 0$, and read every block $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$, $n \le N$, in its occupation basis listed in an order fixed for that block. Enclosures are complex enclosures (Definition 9.1), and "the rules" are those of Lemma 9.2 (i)–(iv).

(i) *Port Grams of a compiled receiver.* Suppose that the instrument matrix of $\mathfrak{R}$ is the compiled instrument matrix $G = D_\eta G^\sigma_{\mathfrak{E}}$ of a well-formed graph for a linear extension σ (Definition 3.3), and that enclosures are given of the matrix $\hat{G}_e$ of every element e and of $\sqrt{\eta_k}$ for every port k. Enclose every other entry of D_η, and every entry of each $E_{\{k\}}$, exactly by itself. Then the rules, applied to the products (3.1) in the order of σ, to the restriction of G to the columns $\mathbb{I}$, and to (2.2), give enclosures of the A_k.

(ii) *Blocks.* Let (C_k, r_k) be enclosures of the A_k, and for every $n \le N$ and occupation vectors $\boldsymbol{m}, \boldsymbol{m}'$ of weight n let $(a_{\boldsymbol{m}\boldsymbol{m}'}, e_{\boldsymbol{m}\boldsymbol{m}'})$ be a square-root enclosure of $1/(\boldsymbol{m}!\,\boldsymbol{m}'!)$. For $J \subseteq [K]$ enclose $\mathbb{1} - A_J$ by the rules from the (C_k, r_k), the identity being exact; enclose the entry $(\boldsymbol{m}, \boldsymbol{m}')$ of $\Pi_{\mathrm{vac}}(J)^{(n)}$ by the rules applied to the product of per $\big((\mathbb{1} - A_J)[\boldsymbol{m}|\boldsymbol{m}']\big)$ with $\sqrt{1/(\boldsymbol{m}!\,\boldsymbol{m}'!)}$; enclose $M_S^{(=n)}$ by the rules applied to (4.7), the signs being exact; and enclose $F_v^{(=n)}$ by the rules applied to $\sum_{S\in\varpi^{-1}(v)} M_S^{(=n)}$. Each enclosure so obtained is a complex enclosure of the block it names.

(iii) *The reference.* Replace each enclosure of a block $F_v^{(=n)}$ of (ii) by the one with a Hermitian centre of Lemma 9.2 (v). Let $\mathfrak{m}_N$ be a sector map (Definition "Sector maps and block-list operators" (foundations/def:sector-map)) whose blocks are the photon numbers $n = 0, \dots, N$ of the declared inputs, in increasing order, of dimensions $\binom{|\mathbb{I}|+n-1}{n}$, with any block kind and block identifiers, each block read in its fixed order. Then the family $\{F_v\}$ restricted to the blocks $n \le N$, read with these enclosures, is a reference with constant 1 against $\mathfrak{m}_N$ over $\mathcal{V}$ (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)); the same holds for $\{M_S\}$ over the click patterns.

(iv) *Exactness.* An entry of an enclosure of (i)–(iii) is exact where every enclosure it is computed from is exact; the reference of (iii) is an exact reference when every one of its radii is zero.

Definition 9.6 (Port dilation of a receiver instrument)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}_{k=1}^K$ on $\mathbb{C}^{\mathbb{I}}$ (Definitions 2.1 and 2.2). A *port dilation* of $\mathfrak{R}$ is a triple $(\mathbb{O}; \mathbb{O}_1, \dots, \mathbb{O}_K; \Omega)$ of a finite mode set $\mathbb{O}$ with a fixed order, pairwise disjoint subsets $\mathbb{O}_1, \dots, \mathbb{O}_K$ of $\mathbb{O}$, its *ports*, and a linear map $\Omega : \mathbb{C}^{\mathbb{I}} \to \mathbb{C}^{\mathbb{O}}$ with

$$\Omega^\dagger \Omega = \mathbb{1}_{\mathbb{I}} \qquad \text{and} \qquad \Omega^\dagger E_{\mathbb{O}_k} \Omega = A_k \quad (k = 1, \dots, K), \tag{9.3}$$

where $E_{\mathbb{O}_k}$ is the coordinate projector of $\mathbb{C}^{\mathbb{O}}$ onto the modes of $\mathbb{O}_k$. The modes of $\mathbb{O}$ in no port are its *loss modes*, and the *row* of a mode $i \in \mathbb{O}$ is the row of Ω that i indexes. The *click pattern* of an occupation vector $\boldsymbol{m} \in \mathbb{N}_0^{\mathbb{O}}$ (Definition 1.2) is

$$S_{\boldsymbol{m}} \coloneqq \big\{ k \in [K] : \; m_i \ge 1 \text{ for some } i \in \mathbb{O}_k \big\}. \tag{9.4}$$

Theorem 9.7 (Photon-number blocks in enclosed factor form)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}_{k=1}^K$ on $\mathbb{C}^{\mathbb{I}}$ and click-pattern POVM $\{M_S\}_{S \subseteq [K]}$ (Theorem 4.5), and let $(\mathbb{O}; \mathbb{O}_1, \dots, \mathbb{O}_K; \Omega)$ be a port dilation of $\mathfrak{R}$ (Definition 9.6), with $S_{\boldsymbol{m}}$ the click pattern of an occupation vector $\boldsymbol{m}$ of $\mathbb{O}$. Read every block $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$ in its occupation basis listed in an order fixed for that block, and for an occupation vector $\boldsymbol{m} \in \mathbb{N}_0^{\mathbb{O}}$ of weight n put

$$c_{\boldsymbol{m}} \coloneqq \mathfrak{F}(\Omega)^\dagger \, |\boldsymbol{m}\rangle \;\in\; \mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}}). \tag{9.5}$$

Enclosures are complex enclosures (Definition 9.1), and "the rules" are those of Lemma 9.2 (i)–(iv).

(i) *The factor form.* For every $n \ge 0$ and all reals $w_S \ge 0$, $S \subseteq [K]$,

$$\sum_{S \subseteq [K]} w_S \, M_S^{(=n)} \;=\; \sum_{\boldsymbol{m} :\, |\boldsymbol{m}| = n} w_{S_{\boldsymbol{m}}} \, c_{\boldsymbol{m}} c_{\boldsymbol{m}}^\dagger, \tag{9.6}$$

the sum on the right over the occupation vectors of $\mathbb{O}$ of weight n. In particular $M_S^{(=n)}$ is the sum of the $c_{\boldsymbol{m}} c_{\boldsymbol{m}}^\dagger$ with $S_{\boldsymbol{m}} = S$, and the block $F_v^{(=n)}$ of the emitted POVM of a declared coarse-graining ϖ (Definition 5.5) is (9.6) with $w_S = 1$ for $S \in \varpi^{-1}(v)$ and $w_S = 0$ otherwise. The entry of $c_{\boldsymbol{m}}$ at an occupation vector $\boldsymbol{m}'$ of $\mathbb{I}$ is $\operatorname{per}\Omega[\boldsymbol{m}|\boldsymbol{m}']/\sqrt{\boldsymbol{m}!\,\boldsymbol{m}'!}$, with $\Omega[\boldsymbol{m}|\boldsymbol{m}']$ as in Lemma 9.4 (ii).

(ii) *Enclosed columns.* Let enclosures of the entries of Ω be given, and, for all occupation vectors $\boldsymbol{m}$ of $\mathbb{O}$ and $\boldsymbol{m}'$ of $\mathbb{I}$ of weight n, a square-root enclosure of $1/(\boldsymbol{m}!\,\boldsymbol{m}'!)$. The rules, applied to the conjugate of $\operatorname{per}\Omega[\boldsymbol{m}|\boldsymbol{m}']$ times that square root, enclose every entry of every $c_{\boldsymbol{m}}$. An entry so enclosed is exact where every enclosure it is computed from is exact, and it is enclosed exactly as 0 where every term of the permanent, a product of entries of Ω, has a factor enclosed exactly as 0; in particular every entry of $c_{\boldsymbol{m}}$ is, where $\boldsymbol{m}$ has a photon in a mode whose row is enclosed exactly as zero.

(iii) *The enclosure of the factor form.* Let $n \geq 0$, let the weights $w_S \geq 0$ be rationals, let enclosures of the columns $c_{\boldsymbol{m}}$ be given, such as those of (ii), and let $\mathcal{C}$ be the set of the occupation vectors $\boldsymbol{m}$ of $\mathbb{O}$ of weight n with $w_{S_{\boldsymbol{m}}} > 0$ whose column is not enclosed exactly as zero, listed in a fixed order. Let (L_c, ϱ) be the matrix box of complex $\dim \operatorname{Sym}^n(\mathbb{C}^{\mathbb{I}}) \times |\mathcal{C}|$ matrices whose column $\boldsymbol{m}$ has the centres and the radii of the enclosure of $c_{\boldsymbol{m}}$, and $W := \operatorname{diag}(w_{S_{\boldsymbol{m}}})_{\boldsymbol{m}\in\mathcal{C}}$. Then (L_c, ϱ, W) is a factor enclosure (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)) of the block (9.6), with $|\mathcal{C}|$ columns; where $\mathcal{C}$ is empty, that block is zero.

(iv) *Port dilations from factors.* Let $\mathbb{O}_0, \mathbb{O}_1, \ldots, \mathbb{O}_K$ be pairwise disjoint finite mode sets and $Y_k : \mathbb{C}^{\mathbb{I}} \to \mathbb{C}^{\mathbb{O}_k}$ linear maps with $Y_k^\dagger Y_k = A_k$ for $k = 1, \ldots, K$ and $Y_0^\dagger Y_0 = \mathbb{L}$, the loss deficiency (Definition 2.2). Then the map Ω into $\mathbb{O} := \mathbb{O}_0 \sqcup \mathbb{O}_1 \sqcup \cdots \sqcup \mathbb{O}_K$, in any fixed order, whose rows on $\mathbb{O}_k$ are those of Y_k for $k = 0, \ldots, K$, is with the ports $\mathbb{O}_1, \ldots, \mathbb{O}_K$ a port dilation of $\mathfrak{R}$. If a matrix X is $E^\dagger D E$ with D diagonal with positive rational diagonal entries d_i and E with entries in $\mathbb{Q}(i)$, then $Y := D^{1/2}E$ satisfies $Y^\dagger Y = X$, and the rules applied to the products of square-root enclosures of the d_i with the entries of E enclose Y, exactly in every row i in which d_i is the square of a rational.

Definition 9.8 (Gaussian-rational port Grams and the exact route)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}_{k=1}^K$ on $\mathbb{C}^{\mathbb{I}}$ (Definitions 2.1 and 2.2), and write $\mathbb{Q}(i) = \{x + iy : x, y \in \mathbb{Q}\}$. A declaration of a receiver is

on the *exact route* when (i) the port Grams $\{A_k\}$ that its declared data determine, computed in exact arithmetic, have entries in $\mathbb{Q}(i)$, are Hermitian, and are an admissible Gram family exactly (Definition 2.2), that is,

$$A_k \succeq 0 \quad (k = 1, \dots, K) \qquad \text{and} \qquad \mathbb{1} - A_{[K]} \succeq 0, \tag{9.7}$$

conditions on Gaussian-rational matrices; and (ii) its probe declaration, where it has one, is of the Gaussian tier (Definition 2.3), whose mean photon numbers $\bar{n}_m^{(t)}$ are declared values and so rationals (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)). Under (i) the port Grams are those of a receiver instrument (Theorem 5.1 (ii)), and every receiver instrument with these port Grams induces the same click-pattern POVM (Theorem 5.1 (i)); where the instrument matrix G, declared or compiled, satisfies $G^\dagger G \preceq \mathbb{1}$ exactly, the declared instrument is one of them. A declared family admitted only at the receiver tolerances (Definition 10.4) and not admissible exactly is not on the exact route. A port Gram family or an instrument matrix declared directly has entries that are dyadic rationals (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)); its port Grams then have entries in $\mathbb{Q}(i)$, by (2.2) for a declared instrument matrix, and such a declaration satisfies (i) exactly when its port Grams are Hermitian, as they always are for a declared instrument matrix, and admissible exactly.

Theorem 9.9 (Exact entries of the blocks on the exact route)**.** Let $\mathfrak{R}$ be a receiver instrument whose port Grams $\{A_k\}$ have entries in $\mathbb{Q}(i)$, as on the exact route (Definition 9.8), with click-pattern POVM $\{M_S\}$, a declared coarse-graining ϖ and emitted POVM $\{F_v\}$. Let $n \geq 0$ and let $\boldsymbol{m}, \boldsymbol{m}'$ be occupation vectors of weight n such that $\boldsymbol{m}!\,\boldsymbol{m}'!$ is the square of an integer; this holds on the diagonal, $\boldsymbol{m} = \boldsymbol{m}'$, and for every $\boldsymbol{m}, \boldsymbol{m}'$ when $n \leq 1$.

(i) The entry $(\boldsymbol{m}, \boldsymbol{m}')$ of every block $\Pi_{\mathrm{vac}}(J)^{(n)}$, $M_S^{(=n)}$ and $F_v^{(=n)}$ lies in $\mathbb{Q}(i)$.

(ii) If the enclosures of the A_k in Theorem 9.5 (ii) are exact, that is, each A_k is enclosed by itself with radius zero, then the enclosure of the entry $(\boldsymbol{m}, \boldsymbol{m}')$ of each of these blocks in Theorem 9.5 (ii), and of the blocks $F_v^{(=n)}$ in Theorem 9.5 (iii), is exact, its centre being the entry.

(iii) Under the hypothesis of (ii), every block $n \leq 1$ of the reference of Theorem 9.5 (iii) has radius zero everywhere, and for a cutoff $N \leq 1$ that reference is an exact reference (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)).

10 Residuals, tolerances and floating-point evaluation

Definition 10.1 (The six residuals)**.** The *residuals* of an evaluation of this contract are those of the six quantities below that exist on the input route taken and for the evaluation requested (which exist is stated after (10.7)):

$$r_1 \coloneqq \left\| \left(G^{\dagger}G - \mathbb{1}\right)_{+} \right\| \qquad \text{(contractivity defect: the positive p} \tag{10.1}$$

$$r_2 \coloneqq \max_{k} \ \max\left\{0,\ -\lambda_{\min}(A_k)\right\} \qquad \text{(Gram positivity),} \tag{10.2}$$

$$r_3 \coloneqq \left\| \mathbb{1}_{\mathbb{I}} - A_{\Sigma} - G_{\mathbb{I}}^{\dagger} E_{\mathrm{un}} G_{\mathbb{I}} - \left(\mathbb{1}_{\mathbb{I}} - G_{\mathbb{I}}^{\dagger} G_{\mathbb{I}}\right) \right\| \qquad \text{(deficiency identity (4.3)),} \tag{10.3}$$

$$r_4 \coloneqq \left\| \mathbb{1} - \sum_{S \subseteq [K]} M_{S|t}^{(=n)} \right\| \ \text{ for each emitted } n \text{ and each setting } t \qquad \text{(POVM completeness, blockwise),} \tag{10.4}$$

$$r_5 \coloneqq \max_{v,\,n,\,t} \ \varrho\big(M_{v|t}^{(=n)}\big) \qquad \text{(block Hermiticity and positivity),} \tag{10.5}$$

$$r_6 \coloneqq \max_{t,\ N<n\le m_t^{*}} \left(\beta^{(t)}(n) - \widehat{\beta}^{*(t)}(n)\right)_{+} \qquad \text{(window consistency, Lem. 7.2),} \tag{10.6}$$

in which the *Hermiticity-and-positivity defect* of a square matrix X is

$$\varrho(X) \ \coloneqq \ \max\left\{ \left\|X - X^{\dagger}\right\|,\ \ \left(-\lambda_{\min}\left(\tfrac{1}{2}(X + X^{\dagger})\right)\right)_{+} \right\}, \tag{10.7}$$

where E_{un} projects onto the declared modes in no port, $(\cdot)_{+}$ is the positive part, and $\|\cdot\|$ is the spectral norm throughout (10.1)–(10.7). The maximum in (10.5) runs over exactly the blocks $M_{v|t}^{(=n)}$ of the coarse-grained elements, before any dark-count channel, that the artifact emits: every emitted outcome letter $v \in \mathcal{V}$, every photon number $0 \le n \le N$, and every setting $t \in \mathcal{T}_{\mathrm{set}}$ of the setting family (Definition 8.1; one setting when $\mathbb{I}_P = \varnothing$, Corollary 8.2); the emitted bound of the one-setting family is Corollary 8.15, and that of a setting family a submodule constructs is that submodule's instance of Theorem 8.14. Here $M_{v|t}^{(=n)}$ is the emitted block: the n-photon block of the coarse-grained conditional element $F_{v|t}$, written $F_v^{(=n)}$ in Theorem 5.6 (i) at a single setting.

Which residuals exist on which route. r_2, r_4 and r_5 are functions of the port Grams and of the emitted blocks, which every route produces, so all three exist on all three routes. r_1 (10.1) and r_3 (10.3) are written on G and

$G_{\mathbb{I}}$, and route (i) supplies no instrument matrix: there the Grams are the declared datum and no G exists, so r_1 and r_3 are *not defined*. Synthesizing a G from Theorem 5.1 (ii) to evaluate them on would test the synthesis rather than the declaration. This agrees with Theorem 10.10 (iv), where route (i) is the case in which "there is no compilation". On routes (ii) and (iii) all five of $r_1, \dots, r_5$ exist.

r_6 *depends on the request, not on the route.* It compares the two evaluations of a single proved inequality, $\beta^{(t)}(n) \le \beta^{*(t)}(n)$ (Lemma 7.2), so its exact value is 0 like the other five, and it exists on all three routes but *only at the settings whose declaration carries an exact-window depth* (Definition 8.7), over exactly the indices $N < n \le m_t^*$ that depth names. A declaration that requests no exact window computes no $\widehat{\beta}^*$ and has nothing to compare, so r_6 is not defined there, as r_1 and r_3 are not on route (i).

No residual bounds the distance between a computed quantity and the exact one (Theorem 10.10 (ii)).

Proposition 10.2 (Properties of the Hermiticity-and-positivity defect)**.** Let X be a complex $D \times D$ matrix and $\varrho(X)$ its Hermiticity-and-positivity defect (10.7).

(i) $\varrho(X) = 0$ if and only if $X = X^\dagger$ and $X \succeq 0$.

(ii) If $X = X^\dagger$ then $\varrho(X) = \big(-\lambda_{\min}(X)\big)_+$, which is the form (10.2) takes on a Gram.

(iii) For every unit vector $|\psi\rangle$, $\operatorname{Re}\langle\psi| X |\psi\rangle \ge -\varrho(X)$ and $\big|\operatorname{Im}\langle\psi| X |\psi\rangle\big| \le \frac{1}{2}\varrho(X)$. Hence $\varrho(X) \le \varepsilon$ bounds every diagonal expectation of X away from the negative reals by ε and off the real axis by $\varepsilon/2$.

(iv) Every block $M_{v|t}^{(=n)}$ of the coarse-grained elements, before any dark-count channel, that the artifact emits satisfies $\varrho = 0$ *exactly*: for each v, n and t, $M_{v|t}^{(=n)}$ is a diagonal block of the positive operator $F_{v|t}$ (Definition 8.1 (a) and Theorem 5.6), and a diagonal block — the compression $\Pi X \Pi$ of a positive operator X by an orthogonal projector Π, read on the range of Π — is Hermitian and positive semidefinite. So r_5 is a residual in exactly the sense of the other five: the exact value is 0, and what is emitted is the floating-point evaluation.

Definition 10.4 (Receiver tolerances and the exact conditions they decide)**.** Several conditions this contract states are *exact* statements about declared data that this contract nevertheless decides in floating point. Each is decided by a declared number, fixed here, and by no other threshold. The *receiver tolerances* are the declared $\varepsilon_1, \dots, \varepsilon_6$, ε_{gap} and ε_{eig}, each in the declared

domain $[0, 1)$; every quantitative statement below uses that upper bound and says where. Items (i) to (v) spend ε_1, ε_2 and ε_4; ε_3, ε_5, ε_6 and ε_{gap} decide no exact condition on declared data and bound computed quantities only (kind (c) below); and ε_{eig} decides nothing at all. It is subtracted from a computed eigenvalue in (8.6), so it is the one declared number this contract spends on a *hypothesis* (Assumption 11.11) rather than on a test.

(i) *Route-(i) Gram admissibility*, $A_k \succeq 0$ and $\sum_k A_k \preceq \mathbb{1}$ (Definition 2.2), is decided at ε_2: the family is admitted when $\lambda_{\min}(A_k) \geq -\varepsilon_2$ for every k and $\lambda_{\max}(A_\Sigma) \leq 1+\varepsilon_2$. ε_2 is the tolerance the contract already attaches to Gram positivity through (10.2), so the same declared number bounds the residual and decides the admission, in the same units.

(ii) *The route-(i) rank consistency check* $\operatorname{rank} A_k \leq |\mathbb{P}_k|$ is decided by the ε_2-rank

$$\operatorname{rank}_{\varepsilon_2}(X) \;:=\; \#\big\{\, i :\; \lambda_i\big(\tfrac{1}{2}(X+X^\dagger)\big) > \varepsilon_2 \,\big\} \qquad \text{(eigenvalues with multiplicity)}, \tag{10.8}$$

the check being $\operatorname{rank}_{\varepsilon_2}(A_k) \leq |\mathbb{P}_k|$ for every k. The threshold is *absolute* and it is ε_2; no separate tolerance is added.

(iii) *Dark-channel admissibility* for a *declared* R is decided in the detector-imperfections submodule, Definition 19.3, which extends this list; the tolerances that definition spends are the ε_4 and ε_2 of this list.

(iv) *Bound-data admissibility*, $0 \leq c$ in (6.1), is decided at ε_2: the data are admitted when $c = 1-\lambda_{\max}(B) \geq -\varepsilon_2$. *Nothing is clamped*: the computed c enters (6.2) exactly as it stands. The remaining inequalities of (6.1), $\gamma \leq \beta_k \leq 1$ and $c \leq d_k \leq 1$, are *not* separately tested and no threshold decides them: they are redundant by theorem, discharged by the admissibility paragraph of Theorem 6.2 from (6.3) and from $\gamma \leq \beta_k$, both hypotheses of the setting family (Definition 8.1 (b)), and their floating-point evaluation is governed by Theorem 10.10 (ii) like every other computed quantity.

Item (iv) decides this admission alone, on every input route, and item (v) does not decide it. Whenever $\varepsilon_1 > \varepsilon_2$ the band

$$c \in \big[\, -\varepsilon_1,\; -\varepsilon_2 \big) \tag{10.9}$$

is non-empty, admitted by item (v)'s test and refused by item (iv)'s, and item (iv) wins on it: the band is refused.

(v) *Route-(ii) contractivity, and (E1) at catalogue entry C10.* $G^\dagger G \preceq \mathbb{1}$ (Definition 2.1) is decided at ε_1: the matrix is admitted when $\lambda_{\max}(G^\dagger G) \leq 1 + \varepsilon_1$. The same test decides the (E1) predicate

of the catalogue's generic block entry C10 (Definition 16.5 of the component-catalogue submodule).

The tolerance that bounds the residual r_1 also decides this admission. $G^\dagger G$ is Hermitian positive semidefinite, so $\big(G^\dagger G - \mathbb{1}\big)_+$ has eigenvalues $\big(\lambda_i(G^\dagger G) - 1\big)_+$ and therefore

$$r_1 = \big\|\big(G^\dagger G - \mathbb{1}\big)_+\big\| = \big(\lambda_{\max}(G^\dagger G) - 1\big)_+, \tag{10.10}$$

so "$r_1 \le \varepsilon_1$" and "$\lambda_{\max}(G^\dagger G) \le 1 + \varepsilon_1$" are the *same test on the same number*, and no separate tolerance is added.

The five items, together with the two extensions the submodules declare in the same form, Definition 19.3, which is item (iii)'s content, and Definition 16.5, are the complete set of exact conditions about declared *data that this module and its submodules decide under a threshold.* Every other validation they state is one of three kinds, and none of them adds a threshold this definition has not named:

(a) a comparison of declared data against a declared domain: every parameter domain of Table 4 except C10's, the detector-layer ranges, $m_t > N$, $m_t^* > N$, $m_t^* \le m_t$, $N \ge 0$, the domain of the receiver tolerances itself, the test of the reserved tolerance field of Assumption 11.8, and the diagonal non-negativity of $\Gamma^{(t)}$;

(b) a combinatorial predicate on the declaration decided by finite inspection: (E2) and (E3) of Definition 3.2, route exclusivity, the port map's disjointness, totality and surjectivity of ϖ, the single-valued τ of Definition 1.1, the exclusivity of the two dark-count routes of Definition 19.1, predicate (X) of Definition 8.3, and the conditional presence of $\varepsilon_{\mathrm{eig}}$ and ε_6 at a declaration carrying an exact-window depth;

(c) the residual tests themselves: r_3, r_4, r_5, r_6 against ε_3, ε_4, ε_5, ε_6, and the residual gap g_{N,m_t} against $\varepsilon_{\mathrm{gap}}$. These are threshold decisions, but on *computed* quantities rather than on declared data, and each is a declared-threshold decision, for the residuals of Definition 10.1 and for the residual gap (6.19), so nothing undeclared decides them either. The tests $r_1 \le \varepsilon_1$ and $r_2 \le \varepsilon_2$ are of this kind too, except where they coincide with an item: on route (ii), where G is declared, the test $r_1 \le \varepsilon_1$ *is* item (v) by (10.10), and on route (i), where the Grams are declared, $r_2 \le \varepsilon_2$ is the positivity half of item (i) written as a residual; on route (iii) r_1 tests the compiled G, and on routes (ii) and (iii) r_2 tests computed Grams. Neither is $\varepsilon_{\mathrm{eig}}$, which is not a test of any kind: it is subtracted from $\widehat{\beta}^*$ in (8.6) and changes an emitted value, not an admission.

Definition 10.9 (The evaluation model: floating point, enclosures and rigour)**.** The claims of this contract are exact statements about exactly declared data (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)). This contract *evaluates* them in binary64 floating-point arithmetic (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64)), reports those of the six residuals of Definition 10.1 that exist on the input route taken and for the evaluation requested, and asserts no rigorous enclosure of any quantity it computes in floating point (Assumption 11.11 is the one exception): this is its *floating-point evaluation*. For a declaration whose model is an admissible receiver exactly it moreover computes, in exact rational arithmetic, complex enclosures (Definition 9.1) of the photon-number blocks of the emitted family: those of Theorem 9.5, and in the submodules those of the setting families a submodule constructs, each enclosed by that submodule, and of their composition with the dark counts (Corollary 20.2). They are exact where the declaration is on the exact route (Definition 9.8) and the entry is known exactly (Theorem 9.9). From the same enclosures of the port Grams it computes, in exact rational arithmetic, at every setting at which the rules of Lemma 9.2 give them, complex enclosures of the setting's operators $B^{(t)}$, $B_k^{(t)}$ and prefactors $\gamma^{(t)}$, $\beta_k^{(t)}$ (Definition 8.1) and, at a setting carrying an exact-window depth, of its blocks $M_{\mathrm{mc}|t}^{(=n)}$ in the window; from these, an upper bound u on $\lambda_{\max}(B^{(t)})$ and lower bounds ℓ_k on $\lambda_{\min}(B_k^{(t)})$ and λ_n on the least eigenvalue of each window block, by the least-eigenvalue bounds over an enclosure (Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound), Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure)), and a lower bound g on $\gamma^{(t)}$ and upper bounds b_k on $\beta_k^{(t)}$, each the real part of a centre less or plus its radius; from these, the reals $g, b_k, 1-u, 1-\ell_k$ each intersected with $[0,1]$ and $\max\{0,\lambda_n\}$; and from those, by the formulas of Definition 8.12 (iii), a weight: this is its *rigorous evaluation* of the weights. A weight emitted at a setting is the weight so formed where the rigorous evaluation is formed there, and otherwise the floating-point evaluation of Definition 8.3 or 8.7. These enclosures and bounds are not evaluations, and a decision taken on them is exact; their interval arithmetic is proved in this contract (Lemma 9.2) and their eigenvalue bounds in the foundations contract, and no interval-arithmetic assumption is imported. This *evaluation model* is a *restriction* this contract places on itself and asserts nothing a run could violate.

Theorem 10.10 (Exact statements, floating-point and rigorous evaluation)**.** For the evaluation model of Definition 10.9:

(i) *The asserted mathematics is exact, except where a claim says so in its own statement. Every asserted identity and inequality* of this mod-

ule and its submodules is between exactly specified quantities and contains no error term, no tolerance and no numerical hypothesis (Theorem 6.2, Lemma 6.5, Lemma 6.8, Lemma 7.2 and Lemma 7.4 included, the last two being stated on the exact β^* and carrying no tolerance), *except where a claim's own statement names a receiver tolerance or Assumption 11.11*.

(ii) *The floating-point evaluation is not.* The emitted $\{A_k\}$, block matrices and those bound values that the floating-point evaluation produces are floating-point evaluations of those quantities; the enclosures of item (vi) and the rigorous evaluation of item (vii) are not. This contract asserts no bound on the difference, and the residuals of Definition 10.1 are *diagnostics*: passing them is necessary for the artifact to be self-consistent and is not sufficient for any enclosure statement.

(iii) *The evaluation label.* An artifact carries the rigorous evaluation label exactly where every per-setting weight it emits is formed by the rigorous evaluation of Definition 10.9, as the enclosed weight of Definition 8.12 or a non-negative real below it; every other artifact, on every input route, the route-(iii) compiler included, carries the floating-point evaluation label. The rigorous label concerns the weights alone: the blocks are evaluated in floating point under (ii) whatever the label. No path in this module produces a label stronger than these two.

(iv) *Where route (i) stands.* If the Grams are *declared*, as binary64 values, which are dyadic rationals (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)), the compilation error of (ii) does not exist: there is no compilation. The eigenvalue quantities of Theorem 6.2 are still evaluated in floating point by the floating-point evaluation, so (ii) and (iii) apply to a weight it forms; what changes is that the rigorous evaluation of (vii) then encloses no matrix product, the enclosures it starts from being the declared Grams themselves. This is also why r_1 and r_3 do not exist on this route: they are written on a G that is not supplied (Definition 10.1). Nothing further is claimed.

(v) *The one place a floating-point number is asserted to bound an exact one, and it is opt-in.* Where a setting t carries an exact-window depth and its emitted weight $p_{\mathrm{mc}}^{(t)}$ is that of Definition 8.7, the weight depends on the computed $\widehat{\beta}^*$ through (8.6), and through the minimum so may the uniform p_{mc}; the inequality $\widehat{\beta}^{*(t)}(n) - \varepsilon_{\mathrm{eig}} \le \beta^{*(t)}(n)$ that makes each such emitted value a lower bound is Assumption 11.11, not a theorem of this contract: (ii) governs $\widehat{\beta}^*$ exactly as it governs every other computed quantity, and r_6 is a diagnostic in exactly the

sense (ii) fixes and is blind to the direction the assumption constrains. Where a setting's emitted weight is an enclosed weight, no computed $\widehat{\beta}^*$ and no $\varepsilon_{\mathrm{eig}}$ enters it and Assumption 11.11 is not invoked there, whatever depth is declared. Where no depth is declared, no β^* is computed, Assumption 11.11 is not invoked, and the emitted weights are the values (8.5) defines, expression for expression, evaluated in floating point or on enclosed bound data. Item (iii) covers all of these: an artifact with a weight resting on Assumption 11.11 carries the floating-point label, so no artifact resting on Assumption 11.11 can claim more.

(vi) *The model's blocks are enclosed, and the evaluation approximates them.* For a declaration whose model is an admissible receiver exactly, the photon-number blocks of the click-pattern and emitted POVMs of that model are quantities the identities and inequalities of (i) are about. Their complex enclosures (Definition 9.1) of Theorem 9.5, where they are formed, are computed in exact rational arithmetic and contain them: every entry of a block lies within its radius of its centre, an entry whose radius is zero is its centre, and on the exact route the entries that Theorem 9.9 names have radius zero when the port Grams are enclosed exactly. A positive semidefiniteness or a positive definiteness established through Lemma "Positive semidefiniteness from an entrywise enclosure" (foundations/lem:enclosure-psd) (i) on such an enclosure with a Hermitian centre holds of the block itself. The floating-point block matrices are evaluations of the same blocks and fall under (ii): this contract asserts no bound on their difference from the blocks, so no decision about the blocks is taken on them.

(vii) *The rigorously evaluated weights are bounds.* Let t be a setting of a setting family (Definition 8.1) at which the rigorous evaluation is formed from complex enclosures of $B^{(t)}$, of every $B_k^{(t)}$, of $\gamma^{(t)}$, of every $\beta_k^{(t)}$ and, at a setting carrying a depth, of the window blocks $M_{\mathrm{mc}|t}^{(=n)}$; for a receiver without a probe, whose setting family is that of Corollary 8.2, the enclosures it forms from the enclosures of the port Grams are such. Then the four clipped reals it forms are enclosed bound data of the setting and, at a setting carrying a depth, the window values it forms are rigorous window bounds; the weight it forms from them is their enclosed weight, a rational computed exactly; and it and every real between 0 and it bound the multiclick element of the setting from below on every photon-number block above the cutoff and lie in $[0, 1]$. Its validity rests on no number computed in floating point: such a number enters it, if at all, only as the exact rational it is, of which nothing is assumed, as the approximate eigenbasis of Lemma "Least-eigenvalue lower bound from an approximate eigen-

basis" (foundations/lem:least-eigenvalue-bound) does. So (ii) does not govern it, and it rests on no assumption about arithmetic.

11 Assumptions

Assumption 11.1 [**A-TRUST**] (Trusted and exactly characterised receiver)**.** (i) The declared instrument describes the physical receiver *exactly*: its mode space, its declared element graph or the supplied instrument matrix or port Gram family, the port map, the efficiencies, the dark-count probabilities and the probe states; (ii) the receiver is under the legitimate party's control: it is not selected, tuned or otherwise influenced by the adversary.

Assumption 11.3 [**A-PASSIVE**] (Passive linear optics)**.** (i) Every optical element is passive linear optics: it maps annihilation operators to combinations of annihilation operators, $b_i = \sum_j G_{ij} a_j$, with $G^\dagger G \preceq \mathbb{1}$; (ii) no element mixes a with $a^\dagger$, and none provides gain.

Assumption 11.4 [**A-VAC**] (Vacuum on undeclared input modes)**.** (i) Every mode of $\mathbb{M} \setminus \mathbb{I}$, and every mode of the dilation $\mathbb{E}$, carries the vacuum state at the input; (ii) every mode that can carry a non-vacuum state and can couple to a port is declared in $\mathbb{I}$: in $\mathbb{I}_C$ if the adversary can populate it, in $\mathbb{I}_P$ if the legitimate party does.

Assumption 11.5 [**A-MODES**] (Completeness of the declared modes)**.** (i) Every optical path from outside the receiver to a detector is represented by a declared mode of $\mathbb{M}$.

Assumption 11.6 [**A-THRESH**] (Threshold detectors without memory)**.** (i) Each detector reports one bit per detection gate: click or no click; (ii) there is no photon-number resolution, no dead time, no afterpulsing and no timing jitter; (iii) there is no correlation of any kind between gates or between detectors beyond the declared dark-count channel; (iv) dark counts are independent across rounds, and across ports unless a correlated channel is declared explicitly (Definition 19.1).

Assumption 11.7 [**A-PORT**] (Mode-insensitive port response)**.** (i) Port k clicks if and only if the total photon number in the modes $\mathbb{P}_k$ is nonzero; equivalently, its no-click element is the joint vacuum projector $\Pi_k^0 = \mathfrak{F}(\mathbb{1} - E_{\{k\}})$ on those modes.

Assumption 11.8 [**A-EXACT**] (Exact declared parameters)**.** (i) Every declared component parameter, efficiency, dark-count probability and probe mean photon number is exact.

Assumption 11.9 [**A-NOSEL**] (Device fixed independently of the adversary)**.** (i) The component list, its parameters, the port map and the probe declaration are fixed before the protocol run; (ii) none of them is chosen by the adversary, and none can be influenced by her, in particular not through a degree of freedom the model does not carry, such as wavelength.

Assumption 11.10 [**A-ROUND**] (Round homogeneity of the instrument, probe and dark counts)**.** The family of the instrument, the probe declaration and the dark-count channel is round-homogeneous (Definition "Round homogeneity of a per-round family" (foundations/def:round-homogeneity)) in both clauses of that definition: (i) the same instrument, the same probe declaration and the same dark-count channel describe every round of the protocol; (ii) the receiver acts in every round as they describe, whatever the inputs and outputs of the other rounds, with no memory across rounds.

Assumption 11.11 [**A-EIG**] (Declared slack on the computed block eigenvalues)**.** (i) At every setting t carrying a declared exact-window depth m_t^* whose emitted weight is that of Definition 8.7 formed by the floating-point evaluation of Definition 10.9, and every index $N < n \leq m_t^*$, the value $\widehat{\beta}^{*(t)}(n)$ an implementation computes for the exact block minimum eigenvalue $\beta^{*(t)}(n)$ of (7.1) satisfies

$$\widehat{\beta}^{*(t)}(n) - \varepsilon_{\mathrm{eig}} \;\leq\; \beta^{*(t)}(n). \tag{11.1}$$

Engine requirements

12 Requirements: compilation, representation and outcomes

Requirement 12.1 (Refusal of a graph that is not well formed). The engine refuses a declared element graph (input `element_graph`) that fails any of conditions (E1)–(E3) of Definition 3.2, with the failing condition named, together with the offending element or edge for (E1) and the offending element pair for (E3); no matrix is emitted, and no ordering is invented to repair it. The refusal follows the `fail_closed` convention.

Requirement 12.2 (Admission of Gaussian-tier probe states only). The engine admits a probe declaration (input `probe_declaration`) only in the Gaussian tier of Definition 2.3, whose states are occupation-diagonal by construction, and refuses a declared probe state outside it under the `fail_closed` convention.

Requirement 12.3 (Declared port map). The engine takes the detector ports as a declared input (`port_map`) and never infers them from the instrument matrix: by Remark 4.3, the partition of the output modes into "in some port" and "in no port" has no observable content beyond the port Grams it produces.

Requirement 12.4 (One mode per port only where the rank criterion holds). The engine uses a one-mode-per-port representation of a receiver only where the criterion of Theorem 5.1 (iv), $\operatorname{rank} A_k \leq 1$ for every k, holds, and never assumes it by default. The criterion is decided on the emitted port Grams, that is after every detector efficiency is applied (Remark 18.2), with each rank decided as Requirement 14.3 fixes.

Requirement 12.5 (Declared emitted alphabet and coarse-graining). The engine takes the emitted alphabet $\mathcal{V}$ as an input (`emitted_outcome_alphabet`) and the declared coarse-graining ϖ of Definition 5.5 with it (`outcome_coarse_graining`); it never fixes an outcome alphabet internally. A receiver that fixed six outcomes over four detectors internally would make the outcome alphabet an implementation detail of the receiver rather than a declared property of the protocol, and every downstream index would then depend on an undeclared convention.

13 Requirements: the emitted weight

Requirement 13.1 (Declared window at every setting whose exact flag is general). A declaration supplying no window m_t (`tail_window`) at a setting where predicate (X) of Definition 8.3 fails is refused rather than defaulted; by Remark 8.5 that condition is decidable at input validation from declared data alone. The engine therefore requires m_t exactly at the settings whose exact flag is general, and at no others. A declared window satisfies $m_t > N$ (Definition 8.3); one that does not is refused at every setting, whatever its flag. A window $m_t > N$ supplied at a setting whose flag is exact is unused, not an error.

Requirement 13.2 (Declaration of an exact-window depth and its tolerances). Declaring an exact-window depth (Definition 8.7) at any setting requires ε_{eig} and ε_6 (the bound on r_6 of Definition 10.1); the engine refuses a declaration missing either, rather than defaulting it, exactly as a missing window m_t is refused (Requirement 13.1), with `exact_window_declared_without_required_tolerance`. The engine refuses a depth violating $m_t^* > N$, or violating $m_t^* \leq m_t$ at a setting where predicate (X) of Definition 8.3 fails, with `exact_window_depth_outside_declared_range`.

Requirement 13.3 (Computed block eigenvalues carried as computed). Nothing is clamped, and the computed value is emitted: at every setting carrying an exact-window depth, the engine carries each $\widehat{\beta}^{*(t)}(n)$, $N < n \leq m_t^*$, in the artifact exactly as computed, in `bound_quantifier_domain`, and ε_{eig} enters only as the declared slack subtracted in (8.6) (Definition 8.7), so that a reader can recompute (8.6) and see what the slack cost.

Requirement 13.4 (Provenance of each emitted multiclick weight). The engine evaluates an exact window, that of Definition 8.7 or the rigorous window bounds of Definition 8.12, only where a declaration requests it by an exact-window depth, and never by default. Where it is not requested, the engine may also report block eigenvalues as a diagnostic; the values entering the strengthening slot (Definition 8.16), `high_photon_bounds`, are the per-setting weights $p_{\text{mc}}^{(t)}$ of Theorem 8.14 — at a receiver without a probe the single value p_{mc} — each the enclosed weight of Definition 8.12, rounded down, where Requirement 13.5 forms it, and otherwise the floating-point evaluation of Definition 8.3 or 8.7; and the artifact says, for each, which evaluation produced it, which of Definitions 8.3 and 8.7 gives its formula and at what depth, in `bound_quantifier_domain`.

Requirement 13.5 (Rigorous evaluation of the multiclick weight).

(i) *Where it applies.* On every declaration whose port Grams the engine encloses by Requirement 14.6 (iii), the engine evaluates the weight of every setting by (ii) to (iv), in exact rational arithmetic: the rigorous evaluation of Definition 10.9. At a setting where it cannot, it emits the weight of Definition 8.3 or 8.7 evaluated in floating point, with the reason `port_grams_not_enclosed` where the declaration has no model admissible exactly and `bound_inputs_not_enclosed` where a reciprocal that (ii) forms receives no enclosure.

(ii) *The enclosures.* At each setting t the engine forms, from the enclosures of the port Grams by the rules of Lemma 9.2, complex enclosures of $B^{(t)}$ and of every $B_k^{(t)}$, of $\gamma^{(t)}$ and of every $\beta_k^{(t)}$, and, where t carries an exact-window depth m_t^*, of $M_{\text{mc}|t}^{(=n)}$ for $N < n \leq m_t^*$, each enclosure of one of these Hermitian matrices made Hermitian-centred by part (v) of that lemma. For a receiver without a probe these are, by Corollary 8.2, $B = A_\Sigma$ and $B_k = A_{[K]\setminus\{k\}}$, sums of enclosed Grams, $\gamma = \beta_k = 1$ exactly, and the blocks of (8.3), formed from the enclosures of Theorem 9.5 (ii) taken with m_t^* as its cutoff. For a setting family a submodule constructs they are the enclosures of its prefactors, its operators and its window blocks that the submodule states, formed by the same rules.

(iii) *The four inputs and the window bounds.* For an entrywise enclosure (C, r) of a Hermitian matrix X and a rational $\rho \geq 0$ with $\rho^2 \geq \sum_{i,l} r_{il}^2$, an upper bound on $\lambda_{\max}(C)$ from Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound) (iv) plus ρ bounds $\lambda_{\max}(X)$ from above, and a lower bound on $\lambda_{\min}(C)$ from part (ii) of that lemma less ρ bounds $\lambda_{\min}(X)$ from below (Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (i), (ii)); for a complex enclosure (x_c, r) of a real x, $\operatorname{Re} x_c - r \leq x \leq \operatorname{Re} x_c + r$. By these rules the engine forms an upper bound u on $\lambda_{\max}(B^{(t)})$, lower bounds ℓ_k on $\lambda_{\min}(B_k^{(t)})$, a lower bound g on $\gamma^{(t)}$, upper bounds b_k on $\beta_k^{(t)}$ and lower bounds λ_n on $\lambda_{\min}(M_{\mathrm{mc}|t}^{(=n)})$. Its enclosed bound data are (8.10) of Theorem 8.13 (i), each of the four intersected with $[0, 1]$, and its rigorous window bounds are $\max\{0, \lambda_n\}$. The output of a floating-point eigensolver may enter only as the basis V of Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound), read as the exact rational matrix it is; a basis for which the engine forms no rational $\theta < 1$ satisfying the hypothesis of that lemma is replaced by the identity, which admits $\theta = 0$. No computed eigenvalue is used as a bound, and $\varepsilon_{\mathrm{eig}}$ is not subtracted.

(iv) *The weight.* The engine computes the enclosed weight $\breve{p}_{\mathrm{mc}}^{(t)}$ (Definition 8.12) as an exact rational and emits, as the setting's weight in the strengthening slot `high_photon_bounds` (Definition 8.16), the largest binary64 number at or below it, rounded down as Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction) rounds an emitted lower bound; by Theorem 8.13 (iii) the emitted number bounds the multiclick element of its setting on every block above the cutoff. The uniform weight is the minimum of the emitted per-setting weights.

(v) *What stays.* The floating-point evaluation of the blocks, the residuals and their tests, and, at a setting carrying a depth, the computed values of Requirement 13.3, are formed as they are without this requirement. At a setting evaluated by (ii) to (iv), the floating-point weight of Definition 8.3 or 8.7 enters no emitted number, and it may be recorded beside the emitted one as a diagnostic.

(vi) *The record.* Per setting, in `bound_quantifier_domain`: the evaluation that produced the emitted weight, `rigorous` or `floating_point`, with the reason of (i) where it is `floating_point`; and, where it is `rigorous`, the enclosed bound data, the rigorous window bounds and the enclosed weight as exact rationals, so that a reader can recompute the weight. A setting whose weight is `rigorous` is not one at which

Assumption 11.11 is invoked (Requirement 15.3), and the artifact's evaluation label follows Requirement 14.5.

14 Requirements: residuals, tolerances and the evaluation label

Requirement 14.1 (Emission of the residuals that exist on the route and request). The engine emits with every artifact, in `validation_results`, those residuals of Definition 10.1 that exist on the input route taken and for the evaluation requested, and no other: the keys of r_1 and r_3 are absent, not zero, on route (i), and the key of r_6 is absent, not zero, at a declaration that requests no exact window. The blocks over which (10.5) maximises are the floating-point blocks $M_{v|t}^{(=n)}$ of the coarse-grained elements that Definition 10.1 names, before any dark-count channel, which the `photon_blocks` entry carries beside its dark-composed blocks (Requirement 22.9); r_5 does not run over the dark-composed blocks, on which the exact value 0 of Proposition 10.2 (iv) is not asserted. ε_1 and ε_3 are declared on every route, route (i) included, where they are unused (`numerical_tolerances`); ε_6 is required exactly where an exact-window depth is declared, on the same condition as r_6, and where it is declared elsewhere it is unused and not an error.

Requirement 14.2 (Refusal on a residual above its tolerance, and no repair). The engine compares each residual that exists (Definition 10.1) with its receiver tolerance ε_i, and a residual above its tolerance is a *refusal*, not a warning; in particular $r_5 > \varepsilon_5$ refuses with the reason `photon_block_positivity_residual_above_tolerance`, and $r_6 > \varepsilon_6$ with the reason `window_eigenvalue_consistency_residual_above_tolerance`. In the same way, at every setting where predicate (X) of Definition 8.3 fails, the engine compares the residual gap g_{N,m_t} of (6.19) on the declared window m_t with $\varepsilon_{\mathrm{gap}}$, and a gap above it refuses. The direction is refuse-only: no block is Hermitized, symmetrized, projected onto the positive cone, clamped or repaired in any way; the artifact carries the computed block itself, and r_5 reports that block's own defect. The same holds of r_6: it refuses and repairs nothing (Remark 8.11 states the direction of failure it is blind to).

Requirement 14.3 (Decisions at the receiver tolerances, and refusal of failed bound data). The engine decides each exact condition that Definition 10.4 lists by the test given there and by no other threshold, and every rank statement this module decides on floating-point data by (10.8); no other rule is licensed, there or at Theorem 5.1 (iv). Bound data failing item (iv) of Definition 10.4 are a failed validation, so `fail_closed` applies at this site: no artifact is emitted. The reason string is the implementer's to name, under the rule the refusal rules of the module's interface state for the

validations they do not tabulate, none of the tabulated strings covering this check.

Requirement 14.4 (Validation of the receiver tolerances against their domain). The engine validates each of the receiver tolerances $\varepsilon_1, \dots, \varepsilon_6$, $\varepsilon_{\mathrm{gap}}$ and $\varepsilon_{\mathrm{eig}}$ of Definition 10.4 at input against the declared domain $[0, 1)$, and refuses a value outside that interval with the reason `tolerance_outside_declared_domain`.

Requirement 14.5 (The evaluation label on every artifact). The engine writes `bound_evaluation` on every artifact of this module, on every input route: `rigorous` where the weight of every setting was formed by the rigorous evaluation of Requirement 13.5, and `floating_point` otherwise, an artifact with a weight resting on Assumption 11.11 among them; which evaluation formed each setting's weight is recorded as Requirement 13.5 (vi) says. No path in this module produces another label (Theorem 10.10 (iii)): the label `certified`, which would name a tier and not an evaluation, and the retired fallback reason `receiver_grams_not_enclosed` are never produced. A run whose declaration carries a weight into which a per-setting weight enters that the record of Requirement 13.5 (vi) names as evaluated in floating point, the uniform weight of a slot entry counting the weight of every setting as entering it, is not reported at the certified tier (Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (vii)); in every other respect the tier a run reports from these artifacts follows Requirement "Reporting of the tiers" (foundations/req:tiers).

Requirement 14.6 (Deciding the exact route and enclosing the port Grams).

(i) *Declared Grams or matrix.* On a route that declares the port Grams or the instrument matrix, the engine computes the port Grams in exact rational arithmetic from the declared values, by (2.2) where the instrument matrix is declared, and decides in exact rational arithmetic whether every A_k is Hermitian and whether the family satisfies (9.7), that is, whether it is an admissible Gram family exactly. The declaration has a model admissible exactly, and is then on the exact route (Definition 9.8), exactly when both hold.

(ii) *Component graph.* On the route that declares a component graph, the engine decides in exact rational arithmetic whether the matrix of every generic block, whose entries are declared values, is a contraction, $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}$. Where every one is, the declaration has a model admissible exactly: the compiled instrument matrix is a contraction (Theorem 3.4 (ii)) and its port Grams are an admissible Gram family (Theorem 4.1 (i)), at the exact values of the declared parameters.

For a declaration that has one and whose parameters its exact arithmetic represents, the engine moreover compiles the instrument matrix of Definition 3.3 and the port Grams exactly and decides whether every port Gram has entries in $\mathbb{Q}(i)$, which places the declaration on the exact route. Every transmittance, loss, reflectivity, extinction and efficiency is the rational number the artifact carries for it: the declared value, or, for a loss declared in decibels, the converted efficiency that `presentation_rule` makes the declared datum, a binary64 number and so a dyadic rational (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)). An angle parameter given in the presentation `angle_presentation`, as a rational coefficient c of π, is $c\pi$ exactly; an angle given in radians is a declared value, the dyadic rational it denotes. The arithmetic represents at least the field generated over $\mathbb{Q}(i)$ by $\sqrt{2}$ and by $\sqrt{p}$ and $\sqrt{1-p}$ for every declared rational parameter p in $[0,1]$, which contains every entry of every catalogue entry whose angles are given as coefficients of π, whose axis angles are multiples of $\pi/8$, whose retardances are multiples of $\pi/2$ and whose rotations and phases are multiples of $\pi/4$. A declaration whose parameters the arithmetic does not represent, a non-zero angle given in radians among them, is treated as off the exact route, with its cause recorded by (iv), even where its port Grams happen to have entries in $\mathbb{Q}(i)$ (two half-wave plates in sequence at one such angle, for instance, compose to $-\mathbb{1}$).

(iii) *Enclosure of the port Grams.* For every declaration with a model admissible exactly, the engine computes, in exact rational arithmetic, complex enclosures (Definition 9.1) of its port Grams at the exact values of the declared parameters. On the exact route as decided by (i) or (ii) each port Gram is enclosed exactly, by itself. Otherwise, on the route of (ii), the enclosures are those of Theorem 9.5 (i), with the matrix of every element enclosed by the rules of Lemma 9.2 applied to the formula of its catalogue entry from square-root enclosures and angle enclosures of its parameters, and $\sqrt{\eta_k}$ by a square-root enclosure; where the exact arithmetic of (ii) has computed a port Gram, its entries, which are expressions in square roots of rationals, may be enclosed from those expressions instead. A phase $e^{i\varphi}$ is $\cos\varphi + i\sin\varphi$, and the angle $\delta/2$ of a retarder is again a rational or a rational multiple of π. No floating-point value enters an enclosure.

(iv) *Record.* The `photon_blocks` entry of the artifact records whether (i) or (ii) places the declaration on the exact route and, where neither does, why: a parameter or an angle that the exact arithmetic of (ii) does not represent (a non-zero angle given in radians among them), port Grams that are not Gaussian-rational, or no model admissible exactly, which

is declared port Grams that are not Hermitian, a declared family that is not an admissible Gram family exactly (a declaration admitted only at the receiver tolerances), or a generic block that is not a contraction exactly. Being off the exact route is not a refusal of the receiver and takes nothing away from the floating-point blocks (Definition 10.9); where the declaration has a model admissible exactly, its port Grams are enclosed by (iii).

15 Requirements: the assumptions

REQUIREMENT 15.1 (Validation of the assumptions on the declared data). The engine enforces by validation, wherever it is decidable, every clause of an assumption of this contract that constrains the declared data. The assumptions that the engine never verifies, as the contract says of each (Remark 11.2), form `never_verified_set`, and no code path decides what they say beyond those clauses.

REQUIREMENT 15.2 (Refusal of a non-zero parameter tolerance). The engine validates the reserved `tolerance` field of every declared parameter as absent or exactly zero, and refuses it otherwise.

Component catalogue

16 Catalogue entries and their admissibility

Definition 16.2 (Entries of the component catalogue)**.** The *catalogue entries* are the entry types C1–C10 below. Throughout, a *path* is a spatial index p and carries the two polarization modes (p, H) and (p, V); a matrix written on two paths acts as $\cdot \otimes \mathbb{1}_2$ on polarization unless stated otherwise, and a matrix written on polarization acts on one path.

C1 — Beamsplitter. Support: two paths (p_1, p_2), four modes. Parameters: transmittance $x \in [0, 1]$, reflection phase $\varphi \in [0, 2\pi)$, per-output losses $y_1, y_2 \in [0, 1]$. Core, on the ordered path pair with rows = outputs:

$$B(x,\varphi) = \begin{pmatrix} \sqrt{x} & \sqrt{1-x}\, e^{i\varphi} \\ \sqrt{1-x}\, e^{-i\varphi} & -\sqrt{x} \end{pmatrix}, \qquad G_{\mathrm{C1}} = \Big[\operatorname{diag}\big(\sqrt{1-y_1}, \sqrt{1-y_2}\big)\, B(x,\varphi)\Big] \otimes \mathbb{1}_2. \tag{16.1}$$

A *polarization-dependent beamsplitter* declares (x_H, φ_H) and (x_V, φ_V) separately, with the per-output losses y_1, y_2 of C1, and is diag $\big(\sqrt{1-y_1}, \sqrt{1-y_2}, \sqrt{1-y_1}, \sqrt{1-y_2}\big)\big[B(x_H, B(x_V, \varphi_V)\big]$ in the reordered basis (p_1H, p_2H, p_1V, p_2V); this is the matrix of entry C2 at $(t_H, \varphi_H, t_V, \varphi_V) = (x_H, \varphi_H, x_V, \varphi_V)$ written in that basis.

C2 — Polarizing beamsplitter. Support: two paths, four modes. Parameters: $t_H, t_V \in [0,1]$ (ideal $t_H = 1$, $t_V = 0$), phases $\varphi_H, \varphi_V \in [0, 2\pi)$, losses $y_1, y_2 \in [0,1]$. In the ordering (p_1H, p_1V, p_2H, p_2V),

$$G_{\mathrm{C2}} = \mathrm{diag}\left(\sqrt{1-y_1}, \sqrt{1-y_1}, \sqrt{1-y_2}, \sqrt{1-y_2}\right) \Xi^{\dagger} \left[B(t_H, \varphi_H) \oplus B(t_V, \varphi_V)\right] \Xi, \tag{16.2}$$

where Ξ is the permutation from (p_1H, p_1V, p_2H, p_2V) to (p_1H, p_2H, p_1V, p_2V).

C3 — Retarder*, with half- and quarter-wave presets.* Support: the two polarization modes of one path. Parameters: retardance $\delta \in [0, 2\pi)$, fast-axis angle $\theta \in [0, \pi)$, scalar loss $y \in [0,1]$. With $R(\theta) = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$,

$$W(\delta, \theta) \coloneqq e^{-i\delta/2}\, R(\theta)\, \mathrm{diag}(1, e^{i\delta}) R(\theta)^{\mathsf{T}} = e^{-i\delta/2} \begin{pmatrix} \cos^2\theta + e^{i\delta}\sin^2\theta & (1-e^{i\delta})\sin\theta\cos\theta \\ (1-e^{i\delta})\sin\theta\cos\theta & \sin^2\theta + e^{i\delta}\cos^2\theta \end{pmatrix}, \tag{16.3}$$

and $G_{\mathrm{C3}} = \sqrt{1-y}\, W(\delta, \theta)$. The presets are

$$\mathrm{HWP}(\theta) \coloneqq W(\pi, \theta) = -i \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}, \qquad \mathrm{QWP}(\theta) \coloneqq W(\tfrac{\pi}{2}, \theta), \tag{16.4}$$

so that $\mathrm{HWP}(\pi/8) = -\frac{i}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$.

C4 — Circular retarder, or polarization rotator. Support: the two polarization modes of one path. Parameter: rotation $\rho \in [0, 2\pi)$. $G_{\mathrm{C4}} = R(\rho)$.

C5 — Mirror or fold. Support: one path. Parameters: reflectivities $R_H, R_V \in [0,1]$, phases $\varphi_H, \varphi_V \in [0, 2\pi)$, handedness flip $\mathrm{fl} \in \{\mathrm{false}, \mathrm{true}\}$ (default false).

$$G_{\mathrm{C5}} = \mathrm{diag}\left(\sqrt{R_H}, \sqrt{R_V}\right) \mathrm{diag}\left(e^{i\varphi_H}, e^{i\varphi_V}\right) F, \qquad F = \begin{cases} \mathrm{diag}(1,-1) & \mathrm{fl} = \mathrm{true} \\ \mathbb{1}_2 & \text{otherwise.} \end{cases} \tag{16.5}$$

C6 — Phase shifter. Support: any declared mode subset Σ. Parameters: $\varphi_m \in [0, 2\pi)$ per $m \in \Sigma$. $G_{\mathrm{C6}} = \mathrm{diag}(e^{i\varphi_m})_{m \in \Sigma}$.

C7 — Loss. Support: any declared mode subset Σ. Parameters: $\eta_m \in [0,1]$ per $m \in \Sigma$, optionally declared as $L_{\mathrm{dB}} \geq 0$ with $\eta = 10^{-L_{\mathrm{dB}}/10}$. $G_{\mathrm{C7}} = \mathrm{diag}(\sqrt{\eta_m})_{m \in \Sigma}$.

C8 — Polarizer. Support: the two polarization modes of one path. Parameters: transmission axis $\theta \in [0, \pi)$, pass amplitude $\sqrt{t}$ and leak amplitude $\sqrt{t_\perp}$ with $t, t_\perp \in [0,1]$. With $|e_\theta\rangle = \cos\theta\,|H\rangle + \sin\theta\,|V\rangle$,

$$G_{\mathrm{C8}} = \sqrt{t}\; |e_\theta\rangle\langle e_\theta| + \sqrt{t_\perp}\; \left|e_\theta^\perp\right\rangle\!\left\langle e_\theta^\perp\right| = R(\theta)\, \mathrm{diag}\left(\sqrt{t}, \sqrt{t_\perp}\right) R(\theta)^{\mathsf{T}}. \tag{16.6}$$

C9 — Mode routing. Support: any declared mode subset. Parameter: a permutation π of the modes of Σ; $(G_{\mathrm{C9}})_{ij} = \delta_{i,\pi(j)}$.

C10 — Generic block. Support: any declared mode subset Σ. Parameter: a matrix $G_\Sigma \in \mathbb{C}^{\Sigma \times \Sigma}$, admitted after the explicit check $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}_\Sigma$.

Table 4 collects the supports and the parameter domains.

ID	Support	Parameters (domain)	Lossless exactly when
C1	2 paths (4 modes)	$x \in [0,1]$, $\varphi \in [0,2\pi)$, $y_1, y_2 \in [0,1]$	$y_1 = y_2 = 0$
C2	2 paths (4 modes)	$t_H, t_V \in [0,1]$, $\varphi_H, \varphi_V \in [0,2\pi)$, $y_1, y_2 \in [0,1]$	$y_1 = y_2 = 0$
C3	1 path, polarization	$\delta \in [0,2\pi)$, $\theta \in [0,\pi)$, $y \in [0,1]$	$y = 0$
C4	1 path, polarization	$\rho \in [0,2\pi)$	always
C5	1 path, polarization	$R_H, R_V \in [0,1]$, $\varphi_H, \varphi_V \in [0,2\pi)$, fl	$R_H = R_V = 1$
C6	any Σ	$\varphi_m \in [0,2\pi)$	always
C7	any Σ	$\eta_m \in [0,1]$	every $\eta_m = 1$
C8	1 path, polarization	$\theta \in [0,\pi)$, $t, t_\perp \in [0,1]$	$t = t_\perp = 1$ (then $G_\Sigma = \mathbb{1}$)
C9	any Σ	a permutation of Σ	always
C10	any Σ	G_Σ with $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}$ (checked)	$G_\Sigma^\dagger G_\Sigma = \mathbb{1}$ (checked)

Table 1: The catalogue entries. Support shapes are stated in modes; the parameter domains are the declared domains of Definition 3.2 (E1); the last column names the subfamily on which Theorem 16.3 proves the entry lossless.

Theorem 16.3 (Admissibility of the catalogue entries). Let G_Σ be the matrix of an entry type C1–C10 of Definition 16.2 at a parameter tuple in its declared domain. Then:

(i) $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}_\Sigma$, so G_Σ is a local element in the sense of Definition 1.3;

(ii) G_Σ is lossless, $G_\Sigma^\dagger G_\Sigma = \mathbb{1}_\Sigma$, exactly on the subfamily named in Table 4;

(iii) for C1–C9, a matrix whose parameter tuple passes the domain comparison of (E1) (Definition 3.2) is contractive by (i); C10 is the only entry whose declared domain is the contractivity condition itself, so that its check of Theorem 3.4 (iv)(E1) is a spectral test on the supplied matrix.

Lemma 16.4 (Loss factors of the catalogue entries and the dilation of a graph)**.** For a diagonal matrix D write $D^\flat$ for the matrix of the rows of D whose diagonal entry is not zero, so that $(D^\flat)^\dagger D^\flat = D^\dagger D$.

(i) *Loss factors.* Let G_Σ be the matrix of an entry type of Definition 16.2 at a parameter tuple in its declared domain, in the notation of that definition, and let Θ be the matrix given for its type below, each of its rows indexed by a mode of its own:

- C1: $\Theta = \big(\operatorname{diag}(\sqrt{y_1}, \sqrt{y_2})^\flat\, B(x,\varphi)\big) \otimes \mathbb{1}_2$;

- C2: $\Theta = \operatorname{diag}(\sqrt{y_1}, \sqrt{y_1}, \sqrt{y_2}, \sqrt{y_2})^\flat \, \Xi^\dagger \big[B(t_H, \varphi_H) \oplus B(t_V, \varphi_V)\big] \Xi$;
- C3: $\Theta = \operatorname{diag}(\sqrt{y}, \sqrt{y})^\flat$;
- C4, C6 and C9: Θ with no row;
- C5: $\Theta = \operatorname{diag}\big(\sqrt{1 - R_H}, \sqrt{1 - R_V}\big)^\flat$;
- C7: $\Theta = \operatorname{diag}\big(\sqrt{1 - \eta_m}\big)^\flat_{m \in \Sigma}$;
- C8: $\Theta = \operatorname{diag}\big(\sqrt{1 - t}, \sqrt{1 - t_\perp}\big)^\flat \, R(\theta)^{\mathsf{T}}$;
- C10, with the entries of G_Σ in $\mathbb{Q}(i)$: $\Theta = D^{1/2} E$, where $\mathbb{1}_\Sigma - G_\Sigma^\dagger G_\Sigma = E^\dagger D E$ with D diagonal with positive rational diagonal entries and E with entries in $\mathbb{Q}(i)$; such a decomposition exists, with at most $|\Sigma|$ rows.

Then $G_\Sigma^\dagger G_\Sigma + \Theta^\dagger \Theta = \mathbb{1}_\Sigma$, and Θ is a *loss factor* of the entry. For C1–C9, Θ has no row exactly on the lossless subfamily named in Table 4.

(ii) *The dilation of a compiled graph.* Let $\mathfrak{R}$ be a receiver instrument whose instrument matrix is the compiled instrument matrix $G = D_\eta G_{\mathfrak{E}}^\sigma$ of a well-formed graph $\mathfrak{E}$ of catalogue elements for a linear extension $\sigma = (e_1, \ldots, e_L)$ (Definition 3.3), with ports $\mathbb{P}_1, \ldots, \mathbb{P}_K$ and efficiencies η_k. For each j let Θ_j be a loss factor of (i) of the element e_j, its rows indexed by a mode set $\mathbb{E}_j$, and let $\hat{\Theta}_j : \mathbb{C}^{\mathbb{M}} \to \mathbb{C}^{\mathbb{E}_j}$ act as Θ_j on the modes of the support Σ_{e_j} and as zero on the others. Let $\Theta_\eta := \big(\sum_k \sqrt{1 - \eta_k}\, E_{\{k\}}\big)^\flat$, its rows indexed by a mode set $\mathbb{E}_\eta$, one for each mode of a port with $\eta_k < 1$, the sets $\mathbb{M}$, $\mathbb{E}_\eta$ and the $\mathbb{E}_j$ being pairwise disjoint. With $\iota : \mathbb{C}^{\mathbb{I}} \to \mathbb{C}^{\mathbb{M}}$ the inclusion of the declared input modes, put $X_0 := \iota$ and $X_j := \hat{G}_{e_j} X_{j-1}$ for $j = 1, \ldots, L$, and let

$$\Omega : \mathbb{C}^{\mathbb{I}} \to \mathbb{C}^{\mathbb{M} \sqcup \mathbb{E}_\eta \sqcup \mathbb{E}_L \sqcup \cdots \sqcup \mathbb{E}_1} \tag{16.7}$$

have the rows $D_\eta X_L$ on $\mathbb{M}$, $\Theta_\eta X_L$ on $\mathbb{E}_\eta$ and $\hat{\Theta}_j X_{j-1}$ on $\mathbb{E}_j$. Then Ω, with the ports $\mathbb{P}_1, \ldots, \mathbb{P}_K$ of $\mathfrak{R}$, is a port dilation of $\mathfrak{R}$ (Definition 9.6), and its rows on $\mathbb{M}$ are those of $G_{\mathbb{I}}$.

Definition 16.5 (Admission test of the generic block)**.** The (E1) predicate of catalogue entry C10 of Definition 16.2 is decided by the route-(ii) contractivity test of Definition 10.4 (v) applied to the supplied block G_Σ: the block is admitted when $\lambda_{\max}(G_\Sigma^\dagger G_\Sigma) \le 1 + \varepsilon_1$. C10 is the one entry whose (E1) is a spectral condition rather than a parameter-domain comparison (Theorem 16.3 (iii)). Its lossless sub-case $G_\Sigma^\dagger G_\Sigma = \mathbb{1}$ is an *equality* and is

decided two-sidedly, as $\|G_\Sigma^\dagger G_\Sigma - \mathbb{1}\| \le \varepsilon_1$.

Detector imperfections

18 Detector efficiency

Theorem 18.1 (Detector efficiency in the port Grams). Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}$ and an instrument matrix G that does not yet contain a detector-efficiency layer (for a declared element graph, the product $G_{\mathfrak{E}}^{\sigma}$ of Definition 3.3), let $\eta_1, \dots, \eta_K \in [0,1]$, and let $D_\eta = \sum_k \sqrt{\eta_k}\, E_{\{k\}} + E_{\mathbb{M}\setminus\bigcup_k \mathbb{P}_k}$ be the terminal detector-efficiency layer of Definition 3.3.

(i) *Uniform within a port: a scalar on the Gram.* $D_\eta G$ is a contraction, and its port Grams are

$$A_k \;\longmapsto\; \eta_k\, A_k. \tag{18.1}$$

For $\eta_k > 0$ this preserves $\operatorname{rank} A_k$, so it leaves the one-mode-per-port criterion of Theorem 5.1 (iv) untouched. Every formula of Theorems 4.5, 5.6, 6.2 and 8.14 therefore covers inefficient detectors under the single substitution (18.1); in particular $A_\Sigma \mapsto \sum_k \eta_k A_k$, which is in general not a multiple of A_Σ unless the η_k agree.

(ii) *Non-uniform within a port: not a scalar, and the rank can drop.* If attenuation varies *within* a port, that is, for a diagonal contraction D that is not constant on $\mathbb{P}_k$, the effective Gram is

$$A_k^{\mathrm{eff}} = G_{\mathbb{I}}^\dagger D^\dagger E_{\{k\}} D\, G_{\mathbb{I}}, \tag{18.2}$$

which is in general *not* a multiple of A_k and satisfies only $\operatorname{rank} A_k^{\mathrm{eff}} \le \operatorname{rank} A_k$, with strict inequality possible. All formulas of the theorems named in (i) still apply, to A_k^{eff}.

(iii) *One structural site.* Detector efficiency is a parameter of the terminal layer L1 and enters the compiled instrument matrix $D_\eta G_{\mathfrak{E}}^{\sigma}$ only through the single left factor D_η of Definition 3.3, which is applied after every chain element. It is therefore not an audit but an invariant of the compiler: there is exactly one site at which it can enter.

(iv) *The composition record.* The emitted efficiency record lists every declared loss parameter q_i, under its name, with the element that declares it and its position in the compiled product: the transmittance x of a beamsplitter (C1) and t_H, t_V of a polarizing beamsplitter (C2), which is also the entry of a polarization-dependent beamsplitter, its

x_H, x_V recorded as t_H, t_V; the output losses y_1, y_2 of C1 and C2 and the loss y of a retarder (C3), whose transmittances are $1-y_1$, $1-y_2$ and $1-y$; the reflectivities R_H, R_V of a mirror (C5); the transmittances η_m of a loss element (C7); the pass and leak transmittances $t, t_\perp$ of a polarizer (C8); and the detector efficiencies η_k. Two elements may carry equal numerical values; they are still two losses and appear as two rows.

(v) *The deficiency identity.* The identity (4.3) of Lemma 4.2 holds for every receiver instrument, before and after the layer D_η, and is the correct check that loss was accounted for exactly once: its two summands are the amplitude that left through the dilation and the amplitude that reached an unmonitored declared mode, and neither carries a separate efficiency term; after the layer, the efficiency loss is part of the dilation summand.

19 Dark counts

Definition 19.1 (Dark-count channel and its independent product form)**.** A *dark-count channel* is a matrix $R = (R_{S|S'})_{S,S' \subseteq [K]}$ that is

(D1) *column-stochastic*: $R_{S|S'} \geq 0$ and $\sum_S R_{S|S'} = 1$ for every S';

(D2) *monotone*: $R_{S|S'} > 0$ implies $S' \subseteq S$; a dark count adds a click and never removes one.

The *independent product form* generated by per-port dark-count probabilities $p_{\mathrm{dc}|k} \in [0,1]$ is

$$R_{S|S'} := \mathbb{1}\left[S' \subseteq S\right] \prod_{k \in S \setminus S'} p_{\mathrm{dc}|k} \prod_{k \notin S} \left(1 - p_{\mathrm{dc}|k}\right). \tag{19.1}$$

For $\mathcal{O} \subseteq 2^{[K]}$ we write $R_{\mathcal{O}|S'} := \sum_{S \in \mathcal{O}} R_{S|S'}$. A dark-count channel is supplied on one of two routes: on the *generated dark-count route* it is (19.1), and on the *declared dark-count route* it is a declared matrix R. (D1) and (D2) hold by construction for (19.1) (Theorem 19.2 (i)); for a *declared* R they are conditions to be checked, and in floating point they are decided at the receiver tolerances ε_4 and ε_2 respectively, by item (iii) of Definition 10.4, whose content is Definition 19.3.

Theorem 19.2 (Dark counts on a photon-number block-diagonal POVM). Let R be a dark-count channel, let $\{M_S\}_{S \subseteq [K]}$ be the click-pattern POVM of Theorem 4.5 or, at a setting t of a setting family of K ports, its POVM $\{M_{S|t}\}_{S \subseteq [K]}$ (Definition 8.1 (a)), written $\{M_S\}$, and put $\widetilde{M}_S := \sum_{S' \subseteq [K]} R_{S|S'} M_{S'}$.

(i) *The product form is admissible.* (19.1) satisfies (D1) and (D2).

(ii) *Structure is preserved.* $\{\widetilde{M}_S\}_{S\subseteq[K]}$ is a POVM, every $\widetilde{M}_S$ is block diagonal in the photon number with $\widetilde{M}_S^{(=n)} = \sum_{S'} R_{S|S'} M_{S'}^{(=n)}$, and R acts on the outcome label alone: it commutes with every operation on the optical modes.

(iii) *Multiclick absorption.* (D2) implies $R_{\mathsf{MC}|S'} = 1$ for every $S' \in \mathsf{MC}$: dark counts never move weight out of the multiclick set.

(iv) *The bound improves.* With

$$r_{\min} \coloneqq \min_{S'\in 2^{[K]}\setminus \mathsf{MC}} R_{\mathsf{MC}|S'}, \tag{19.2}$$

the minimum running over *every* non-multiclick column, the no-click column $S' = \varnothing$ included, one has, writing $\widetilde{M}_{\mathsf{MC}} = \sum_{S\in\mathsf{MC}} \widetilde{M}_S$,

$$\widetilde{M}_{\mathsf{MC}} \;\succeq\; (1 - r_{\min})\, M_{\mathsf{MC}} + r_{\min}\, \mathbb{1}, \tag{19.3}$$

and consequently, if $M_{\mathsf{MC}}^{(>N)} \succeq p_{\mathrm{mc}}\, \mathbb{1}^{(>N)}$, then

$$\widetilde{M}_{\mathsf{MC}}^{(>N)} \;\succeq\; \big((1 - r_{\min})\, p_{\mathrm{mc}} + r_{\min}\big)\, \mathbb{1}^{(>N)}, \tag{19.4}$$

which is $\geq p_{\mathrm{mc}}$ since $p_{\mathrm{mc}} \leq 1$.

Definition 19.3 (Receiver tolerances deciding dark-channel admissibility)**.** This definition extends the list of Definition 10.4 by its item (iii), under the same rule: an exact condition about declared data is decided in floating point by a receiver tolerance, fixed here, and by no other threshold.

Dark-channel admissibility (Definition 19.1) for a *declared* R: (D1) column-stochasticity is decided at ε_4, admitted when $R_{S|S'} \geq -\varepsilon_4$ for every entry and $\big|\sum_S R_{S|S'} - 1\big| \leq \varepsilon_4$ for every column; and (D2) monotonicity is decided at ε_2: an entry counts as positive when $R_{S|S'} > \varepsilon_2$, so the channel is refused exactly when some entry with $S' \not\subseteq S$ exceeds ε_2. ε_4 bounds the completeness residual (10.4), which is the same kind of quantity as a column sum's defect from one; ε_2 is the threshold at which this contract already calls a spectral quantity zero.

Proposition 19.4 (Multiclick bound under a dark-count channel admitted with slack)**.** Let $\{M_S\}_{S\subseteq[K]}$ be the click-pattern POVM of Theorem 4.5 or, at a setting t of a setting family of K ports, its POVM $\{M_{S|t}\}_{S\subseteq[K]}$ (Definition 8.1 (a)), written $\{M_S\}$, and for a matrix $R = (R_{S|S'})_{S,S'\subseteq[K]}$ let $\widetilde{M}_S$, $\widetilde{M}_{\mathsf{MC}}$ and $r_{\min}$ be formed as in Theorem 19.2 and (19.2). Put

$$\delta_{\mathrm{dc}} \coloneqq \begin{cases} 0 & \text{generated route: } R \text{ is the product form (19.1),} \\ \varepsilon_4 + (K+1)\,\varepsilon_2 & \text{declared route: } R \text{ admitted under Def. 10.4 (iii),} \end{cases} \tag{19.7}$$

the two cases being the two routes of Definition 19.1, and

$$\bar{r} \;\coloneqq\; \min\{\, r_{\min},\, 1 \,\}. \tag{19.8}$$

Let R be declared and admitted under item (iii) of Definition 10.4 (Definition 19.3), on either side of either test and whatever the sign of $r_{\min}$. Then:

(i) *(the multiclick columns)* for every $S' \in \mathsf{MC}$,

$$R_{\mathsf{MC}|S'} \;\geq\; (1-\varepsilon_4) - (K+1)\varepsilon_2 \;=\; 1-\delta_{\mathrm{dc}}; \tag{19.9}$$

(ii) *(the upper side)* for every column S',

$$R_{\mathsf{MC}|S'} \;\leq\; (1+\varepsilon_4) + (K+1)\varepsilon_4 \;=\; 1+(K+2)\varepsilon_4, \tag{19.10}$$

and consequently

$$0 \;\leq\; r_{\min} - \bar{r} \;\leq\; (K+2)\,\varepsilon_4; \tag{19.11}$$

(iii) *(the composed inequality)*

$$\widetilde{M}_{\mathsf{MC}} \;\succeq\; (1-\delta_{\mathrm{dc}})\,M_{\mathsf{MC}} + \bar{r}\,\big(\mathbb{1} - M_{\mathsf{MC}}\big) \;\succeq\; \Big[(1-\bar{r})\,M_{\mathsf{MC}} + \bar{r}\,\mathbb{1}\Big] - \delta_{\mathrm{dc}}\,\mathbb{1}, \tag{19.12}$$

and consequently, whenever $M_{\mathsf{MC}}^{(>N)} \succeq p_{\mathrm{mc}}\,\mathbb{1}^{(>N)}$,

$$\widetilde{M}_{\mathsf{MC}}^{(>N)} \;\succeq\; \big((1-\bar{r})\,p_{\mathrm{mc}} + \bar{r} - \delta_{\mathrm{dc}}\big)\,\mathbb{1}^{(>N)}. \tag{19.13}$$

20 The dark-composed family and the noise floor

Definition 20.1 (The dark-composed family)**.** Let $\mathcal{T}_{\mathrm{set}}$ be a finite non-empty set of settings and, for every $t \in \mathcal{T}_{\mathrm{set}}$, let $\{M_{S|t}\}_{S\subseteq[K]}$ be a POVM on a Fock space $\mathcal{F}(\mathbb{C}^I)$ (Definition 1.2) every element of which commutes with the photon number. Let R be a dark-count channel (Definition 19.1) and ϖ a declared coarse-graining onto $\mathcal{V}$ (Definition 5.5). The *dark-composed family* is

$$\widetilde{M}_{S|t} \coloneqq \sum_{S'\subseteq[K]} R_{S|S'}\,M_{S'|t}, \qquad \widetilde{F}_{v|t} \coloneqq \sum_{S\in\varpi^{-1}(v)} \widetilde{M}_{S|t} = \sum_{S'\subseteq[K]} R_{\varpi^{-1}(v)|S'}\,M_{S'|t}, \tag{20.1}$$

for $S \subseteq [K]$, $v \in \mathcal{V}$ and $t \in \mathcal{T}_{\mathrm{set}}$. Each $\{\widetilde{M}_{S|t}\}_S$ and each $\{\widetilde{F}_{v|t}\}_v$ is a POVM every element of which commutes with the photon number, with $\widetilde{F}_{v|t}^{(=n)} = \sum_{S'} R_{\varpi^{-1}(v)|S'} M_{S'|t}^{(=n)}$: the coefficients are non-negative scalars, they sum to one over S in every column by (D1), and the fibres of ϖ partition $2^{[K]}$.

Corollary 20.2 (The dark-composed reference, enclosed or exact)**.** Let $\{\widetilde{M}_{S|t}\}$ and $\{\widetilde{F}_{v|t}\}$ be the dark-composed family of POVMs $\{M_{S|t}\}$ on $\mathcal{F}(\mathbb{C}^I)$ (Definition 20.1), with a dark-count channel R whose entries are rational.

(i) Suppose that for some $N \geq 0$ every block $M^{(=n)}_{S|t}$ with $n \leq N$, read in the occupation basis of $\mathrm{Sym}^n(\mathbb{C}^I)$ in an order fixed for that block, has a complex enclosure $\big(C^{(n)}_{S|t}, r^{(n)}_{S|t}\big)$ (Definition 9.1). Then

$$\Big(\sum_{S' \subseteq [K]} R_{\varpi^{-1}(v)|S'}\, C^{(n)}_{S'|t},\ \sum_{S' \subseteq [K]} R_{\varpi^{-1}(v)|S'}\, r^{(n)}_{S'|t} \Big) \tag{20.2}$$

is a complex enclosure of $\widetilde{F}^{(=n)}_{v|t}$, exact on every entry on which every $r^{(n)}_{S'|t}$ with $R_{\varpi^{-1}(v)|S'} \neq 0$ is zero; the same holds for $\widetilde{M}^{(=n)}_{S|t}$, with $R_{S|S'}$ in place of $R_{\varpi^{-1}(v)|S'}$.

(ii) Made Hermitian-centred by Lemma 9.2 (v), the enclosures of (i) make $\{\widetilde{F}_{v|t}\}_v$, restricted to the blocks $n \leq N$, a reference with constant 1 (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)) at every setting, against every sector map whose blocks are the photon numbers $n = 0, \ldots, N$ in increasing order, with any block kind and identifiers, each block read in its fixed order; it is an exact reference when every radius is zero. The same holds for $\{\widetilde{M}_{S|t}\}_S$.

(iii) For a receiver instrument with $\mathbb{I}_P = \varnothing$, the hypothesis of (i) holds with $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$, $M_{S|\mathrm{off}} = M_S$ and the enclosures of Theorem 9.5 (ii), at every N.

(iv) R has rational entries on the generated dark-count route, whose $p_{\mathrm{dc}|k}$ are declared values, and on the declared route, whose entries are declared values; a declared value is the dyadic rational it denotes (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)).

Proposition 20.3 (Loss and dark counts make every dark-composed element definite)**.** Let $\{\widetilde{M}_{S|t}\}$ and $\{\widetilde{F}_{v|t}\}$ be the dark-composed family (Definition 20.1) of a setting family $\big(\mathcal{T}_{\mathrm{set}}, \{M_{S|t}\}\big)$ with operators $B^{(t)}$ and prefactors $\gamma^{(t)}$ (Definition 8.1), a dark-count channel R and a declared coarse-graining ϖ, and write $r_S \coloneqq R_{S|\varnothing}$.

(i) *The floor.* For every t, S, v and $n \geq 0$,

$$\widetilde{M}^{(=n)}_{S|t} \succeq r_S\, \gamma^{(t)} \big(1 - \lambda_{\max}(B^{(t)})\big)^n \mathbb{1}, \qquad \widetilde{F}^{(=n)}_{v|t} \succeq R_{\varpi^{-1}(v)|\varnothing}\, \gamma^{(t)} \big(1 - \lambda_{\max}(B^{(t)})\big)^n \mathbb{1}. \tag{20.3}$$

Hence, if $r_S > 0$ for every $S \subseteq [K]$ and $\lambda_{\max}(B^{(t)}) < 1$, every $\widetilde{M}_{S|t}$ and every $\widetilde{F}_{v|t}$ is positive definite on every photon-number block.

(ii) *Dark counts in product form.* On the generated dark-count route, $r_S = \prod_{k\in S} p_{\mathrm{dc}|k} \prod_{k\notin S}(1-p_{\mathrm{dc}|k})$, and $r_S > 0$ for every S if and only if $0 < p_{\mathrm{dc}|k} < 1$ for every k.

(iii) *Loss through the efficiencies.* If the instrument matrix of a receiver instrument is $D_\eta G'$, with G' a contraction and D_η the terminal detector-efficiency layer, then $\lambda_{\max}(A_\Sigma) \le \max_k \eta_k$. For the unconditioned receiver, with $\mathbb{I}_P = \varnothing$, $B = A_\Sigma$ and $\gamma = 1$, so $\eta_k < 1$ and $0 < p_{\mathrm{dc}|k} < 1$ for every k give $\lambda_{\min}\big(\widetilde{M}_S^{(=n)}\big) \ge r_S\,(1-\max_k \eta_k)^n > 0$.

(iv) *Loss is necessary.* If $\lambda_{\max}(B^{(t)}) = 1$, then for every dark-count channel and every $n \ge 1$ the element $\widetilde{M}_{\varnothing|t}^{(=n)} = r_\varnothing\, M_{\varnothing|t}^{(=n)}$ is singular; if $B^{(t)} = \mathbb{1}$ it is zero.

(v) *Dark counts in every port are necessary.* For the unconditioned receiver on the generated route, let $Z := \{k : p_{\mathrm{dc}|k} = 0\}$. Then $\widetilde{M}_S^{(=n)} = 0$ for every S with $|S \cap Z| > n$; in particular $\widetilde{M}_{\{k\}}^{((=0))} = 0$ for every $k \in Z$. If $p_{\mathrm{dc}|k} = 1$, then $\widetilde{M}_S = 0$ for every $S \not\ni k$.

(vi) *Equivalence.* For the unconditioned receiver on the generated route, every $\widetilde{M}_S$ is positive definite on every photon-number block if and only if $\lambda_{\max}(A_\Sigma) < 1$ and $0 < p_{\mathrm{dc}|k} < 1$ for every k.

21 Efficiency normalization

Definition 21.1 (Pure-loss channel on the channel modes)**.** Let $\mathbb{I}_C$ be the channel modes of a receiver instrument, so that the channel system is $\mathcal{H}(C) = \mathcal{F}(\mathbb{C}^{\mathbb{I}_C})$ (Definition 2.1), let $\mathbb{I}'_C$ be a disjoint copy of $\mathbb{I}_C$, with $m \mapsto m'$ the copying bijection, and let $t \in [0,1]$. Put

$$w_t \colon \mathbb{C}^{\mathbb{I}_C} \to \mathbb{C}^{\mathbb{I}_C \sqcup \mathbb{I}'_C}, \qquad w_t\,|m\rangle := \sqrt{t}\;|m\rangle + \sqrt{1-t}\;\big|m'\big\rangle, \tag{21.1}$$

and identify $\mathcal{F}(\mathbb{C}^{\mathbb{I}_C \sqcup \mathbb{I}'_C})$ with $\mathcal{F}(\mathbb{C}^{\mathbb{I}_C}) \otimes \mathcal{F}(\mathbb{C}^{\mathbb{I}'_C})$ by the occupation identification of Lemma 1.7 (ii), which is unitary. The *pure-loss channel* of transmittance t is

$$\Lambda_t^{\mathrm{loss}}(\rho) \;:=\; \mathrm{tr}_{\mathcal{F}(\mathbb{C}^{\mathbb{I}'_C})}\big[\mathfrak{F}(w_t)\,\rho\,\mathfrak{F}(w_t)^\dagger\big], \qquad \rho \in \mathcal{T}(C), \tag{21.2}$$

and its adjoint is

$$(\Lambda_t^{\mathrm{loss}})^\dagger(Y) \;:=\; \mathfrak{F}(w_t)^\dagger\big(Y \otimes \mathbb{1}_{\mathcal{F}(\mathbb{C}^{\mathbb{I}'_C})}\big)\mathfrak{F}(w_t), \qquad Y \in \mathcal{L}(C). \tag{21.3}$$

Since $w_t^\dagger w_t = t\,\mathbb{1} + (1-t)\,\mathbb{1} = \mathbb{1}$, the map $\mathfrak{F}(w_t)$ is an isometry (Lemma 1.7 (i)); so $\Lambda_t^{\mathrm{loss}}$ is completely positive and trace preserving, $(\Lambda_t^{\mathrm{loss}})^\dagger$ is completely

positive and unital, and $\operatorname{tr}\big[Y\,\Lambda_t^{\mathrm{loss}}(\rho)\big] = \operatorname{tr}\big[(\Lambda_t^{\mathrm{loss}})^\dagger(Y)\,\rho\big]$ for every bounded Y and trace-class ρ, by the defining property of the partial trace and cyclicity of the trace. At $t = 1$ the channel is the identity.

Definition 21.2 (Normalized station and factored loss)**.** Let $\mathfrak{R}$ be a receiver instrument without declared source modes, $\mathbb{I}_P = \varnothing$, with K ports, whose port Grams are

$$A_k = \eta_k\, A_k^\circ, \qquad k = 1, \dots, K, \tag{21.4}$$

for an admissible Gram family $\{A_k^\circ\}$ on $\mathbb{C}^{\mathbb{I}_C}$ and efficiencies $\eta_k \in [0,1]$ with $\eta_{\max} \coloneqq \max_k \eta_k > 0$. This is the case for an instrument matrix $D_\eta G_0$, G_0 a contraction, with A_k° the port Grams of G_0 (Theorem 18.1 (i)), and for a declared Gram family $\{A_k^\circ\}$ with declared efficiencies. For $t \in [\eta_{\max}, 1]$, a *normalized station* of $\mathfrak{R}$ at the *factored loss* t is a receiver instrument $\mathfrak{R}'$ without declared source modes, with the channel modes and the K ports of $\mathfrak{R}$, whose port Grams are

$$A_k' = \frac{\eta_k}{t}\, A_k^\circ, \qquad \text{so that} \qquad A_k = t\, A_k'. \tag{21.5}$$

One exists: $\eta_k/t \in [0,1]$, so $0 \preceq A_k' \preceq A_k^\circ$ and $\sum_k A_k' \preceq \sum_k A_k^\circ \preceq \mathbb{1}$, and an admissible family is the port-Gram family of some receiver instrument (Theorem 5.1 (ii)); for the instrument matrix $D_\eta G_0$ one is $D_{\eta/t} G_0$, a product of two contractions (Proposition 1.4 (ii)), with $D_{\eta/t}$ the terminal layer of Definition 3.3 at the efficiencies η_k/t. Its click-pattern POVM depends on the port Grams alone (Theorem 5.1 (i)), so it is one POVM for every choice. The choice $t = \eta_{\max}$ is the *full* normalization, at which the largest efficiency becomes one, and $t = 1$ gives $\mathfrak{R}'$ the port Grams of $\mathfrak{R}$.

Proposition 21.4 (A receiver as uniform loss followed by its normalized station)**.** Let $\mathfrak{R}$ be a receiver instrument without declared source modes, with efficiencies η_k, $\eta_{\max} > 0$, as in Definition 21.2, let $t \in [\eta_{\max}, 1]$ and let $\mathfrak{R}'$ be a normalized station of $\mathfrak{R}$ at the factored loss t. Let $\{M_S\}$ and $\{M_S'\}$, $S \subseteq [K]$, be their click-pattern POVMs on $\mathcal{H}(C) = \mathcal{F}(\mathbb{C}^{\mathbb{I}_C})$ (Theorem 4.5).

(i) *The click patterns.* For every $S \subseteq [K]$,

$$M_S \;=\; (\Lambda_t^{\mathrm{loss}})^\dagger\big(M_S'\big), \qquad \text{equivalently} \qquad \operatorname{tr}[M_S\,\rho] = \operatorname{tr}\big[M_S'\,\Lambda_t^{\mathrm{loss}}(\rho)\big] \ \text{ for every } \rho \in \mathcal{T}(C). \tag{21.7}$$

(ii) *Classical post-processing, dark counts included.* For every column-stochastic matrix R on $2^{[K]}$, in particular every dark-count channel, and every declared coarse-graining ϖ onto $\mathcal{V}$,

$$\sum_{S'} R_{S|S'} M_{S'} = (\Lambda_t^{\mathrm{loss}})^\dagger\Big(\sum_{S'} R_{S|S'} M_{S'}'\Big), \qquad \sum_{S'} R_{\varpi^{-1}(v)|S'} M_{S'} = (\Lambda_t^{\mathrm{loss}})^\dagger\Big(\sum_{S'} R_{\varpi^{-1}(v)|S'} M_{S'}'\Big) \tag{21.8}$$

so the dark-composed family of $\mathfrak{R}$ with R and ϖ (Definition 20.1) is the pull-back of that of $\mathfrak{R}'$ with the same R and ϖ: the dark counts act after the factored loss.

(iii) *The device.* Under the receiver's device assumptions, Assumptions 11.1, 11.3, 11.4, 11.5, 11.6, 11.7, 11.8, 11.9 and 11.10, $\{M_S\}$ is the measurement the physical receiver makes on the light it receives in its channel modes, and, by Assumptions 11.5 and 11.7, every port responds alike to every mode through which light reaches it; the physical receiver, followed by any dark-count channel, therefore acts on that light exactly as $\Lambda_t^{\mathrm{loss}}$ followed by $\mathfrak{R}'$ and the same dark-count channel.

22 Requirements

Requirement 22.1 (Detector efficiency and component loss in separate fields). A C7 element is not a detector efficiency, and the engine's input schema keeps the two in different fields.

Requirement 22.2 (The efficiency record lists every declared loss parameter). The emitted `efficiency_record` lists every declared loss parameter q_i that Theorem 18.1 (iv) names, by catalogue entry and parameter, with the element that declares it and its position in the compiled product of Definition 3.3. Every row names its parameter beside its value, which is read as that parameter: a transmittance, a reflectivity, or, for y, y_1 and y_2, a loss whose transmittance is $1 - q_i$. Two elements may carry equal numerical values; they are still two losses and appear as two rows. A polarizer's pass transmittance t is one such row, under its element; the factored loss t that the same record carries by Requirement 22.5 is not a row and is recorded under its own name.

Requirement 22.3 (Declared efficiency convention of the port Grams). Which of the two conventions a declaration means, Grams from power data rescaled by η_k or Grams from click data with every η_k declared as 1, is a declared field, and the `efficiency_record` (Requirement 22.2) is where the answer is written down.

Requirement 22.4 (Compilation of a probe-free receiver at its normalized station). For a request whose receiver declares no source modes and some efficiency $\eta_k > 0$, the engine compiles, in place of the declared receiver, its normalized station at the factored loss t, with the efficiency of port k equal to η_k/t, and emits that station's blocks, multiclick weights and dark-composed family; t is the value of `efficiency_normalization` where the request declares one; otherwise, with $\eta_{\max}$ the largest declared

efficiency, $t = \eta_{\max}/(99/100)$, formed exactly, where $\eta_{\max} < 99/100$, and $t = 1$ where $\eta_{\max} \geq 99/100$. On every route the station compiled is the one at the exact rational efficiencies η_k/t: its exact statements, its reference and its enclosures are formed from those rationals, its floating-point evaluation evaluates them as the evaluation model evaluates every declared number, and no value rounded from η_k/t is declared in their place; the quotient is exactly one at a port with $\eta_k = t$. A declared t outside $[\eta_{\max}, 1]$ is refused with `efficiency_normalization_out_of_range`. A request that declares source modes, or whose efficiencies are all zero, is compiled at $t = 1$, and a declared $t < 1$ in such a request is refused with the same reason.

Requirement 22.5 (The efficiency record names the normalization). The emitted `efficiency_record` carries the factored loss t, whether t was declared in the request or set by Requirement 22.4, every declared efficiency η_k beside the exact rational η_k/t at which the receiver was compiled, and whether the normalization was applied, which it was exactly where $t < 1$. Where it was not, it carries the reason, the first that holds of: source modes declared, every efficiency zero, or the factor one, declared or by default. Where $t < 1$ the artefact states that its blocks, multiclick weights and dark-composed family are those of the normalized station at t, whose pull-back through the pure-loss channel of transmittance t is the receiver's measurement.

Requirement 22.6 (The factored loss is applied before the artefact is evaluated). Where a receiver artefact's `efficiency_record` carries a factored loss $t < 1$, every evaluation of its emitted blocks or dark-composed family, or of a target formed from them, on a state that stands for the receiver's statistics on an input ρ of the channel system (an honest-statistics provider forming the honest law among them) is made on $\Lambda_t^{\mathrm{loss}}(\rho)$, or on the state the same construction forms from $\Lambda_t^{\mathrm{loss}}(\rho)$, in place of ρ, with the t of that receiver's record. Where the input is a joint state of which that receiver's channel system is one factor, such as a state shared with another party or with a second normalized receiver, $\Lambda_t^{\mathrm{loss}}$ stands here for $\mathrm{id} \otimes \Lambda_t^{\mathrm{loss}}$ acting on that receiver's channel-system factor, each receiver with the t of its own record. The states the key-rate program optimises over are not inputs in this sense, and the loss is not applied to them. A consumer that cannot apply $\Lambda_t^{\mathrm{loss}}$ to the input it is handed refuses with `factored_loss_not_applied`.

Requirement 22.7 (Refusal of a dark-count channel supplied on both routes). The two routes of Definition 19.1 are exclusive: a request declaring R while some $p_{\mathrm{dc}|k} \neq 0$ is refused with `dark_count_channel_declared_twice`.

Requirement 22.8 (The emitted dark-count channel and the scalars of its composed bound). The artifact's `dark_count_channel` entry carries R, which of the two routes of Definition 19.1 it came from (Remark 19.7), and $r_{\min}$, $\bar{r}$ and δ_{dc} under distinct names. Nothing is clamped: R is emitted as declared and $r_{\min}$ as (19.2) computes it. (19.13), *not* (19.4), is the inequality a downstream module composes with.

Requirement 22.9 (The emitted blocks are dark-composed and carry their reference). (i) The `photon_blocks` entry carries, for every probe setting t and every $n \le N$, the blocks $\widetilde{F}^{(=n)}_{v|t} = \sum_{S' \subseteq [K]} R_{\varpi^{-1}(v)|S'} M^{(=n)}_{S'|t}$ given by the formula (20.1) for the emitted matrix R, in floating point. They are the blocks of the dark-composed family (Definition 20.1) when R is a dark-count channel, satisfying (D1) and (D2) exactly; for a declared R admitted only at the receiver tolerances they are the same formula evaluated at that R.

(ii) For a declaration with a model admissible exactly (Requirement 14.6) whose R satisfies (D1) and (D2) exactly, decided in exact arithmetic on its declared entries on the declared dark-count route, the entry also carries, for every setting and every $n \le N$, a complex enclosure with a Hermitian centre of every block $\widetilde{F}^{(=n)}_{v|t}$ (Definition 9.1), computed in exact rational arithmetic on the exact island (Definition "The exact island" (foundations/def:exact-island) (i)) and with no binary64 value as an input: from the enclosures of the port Grams of Requirement 14.6 (iii), by Theorem 9.5 (ii) for an unconditioned receiver and by the enclosure of its blocks that a submodule constructing the setting family states for that family, then by Corollary 20.2 (i) and Lemma 9.2 (v). The square-root enclosures and angle enclosures used may be any that Definition 9.1 admits. The following entries are enclosed exactly, as zero with radius zero:

(a) every entry of a block $\widetilde{F}^{(=n)}_{v|t}$ such that every $\widetilde{M}^{(=n)}_{S|t}$ with $S \in \varpi^{-1}(v)$, whose sum it is (Definition 20.1), is shown to vanish: for the unconditioned receiver on the generated dark-count route, and for no other receiver or route, by Proposition 20.3 (v), decided on its declared per-port dark-count probabilities; or, for $S = \varnothing$ and $n \ge 1$, by Proposition 20.3 (iv) at a setting whose operator $B^{(t)}$ the engine has computed in exact arithmetic and found to be the identity, which for the unconditioned receiver is A_Σ computed from the port Grams of Requirement 14.6 (i) or (ii);

(b) every entry whose enclosure is exact by Theorem 9.9 (ii) or by the exactness of its blocks that a submodule constructing the setting

family states, carried through Corollary 20.2 (i), and whose value is zero.

No other entry is required to be enclosed as zero.

(iii) The enclosures of (ii), with the family they enclose, are the reference (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)) that the receiver offers for a snap of its emitted family (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference)), one per setting, with constant 1 (Corollary 20.2 (ii)). The entry records the *offered* reference kind of every block: *exact* where every radius of the block is zero, and *enclosed* otherwise, with the largest radius of the block and the largest radius of the enclosures of irrational inputs (square roots, trigonometric values) from which the centres are computed exactly. The record of every snap of the family carries besides the kind of the reference the snap ran on (Requirement "Snapping by the declarer only" (foundations/req:who-snaps)): *exact* on a block whose offered kind is exact, the snap running on its centres, and *none* on every other block, the snap running on the floating-point blocks of (i). The reference is an exact reference where every block is exact. It also names the claims that establish the enclosure, positive semidefiniteness and completeness of the reference, which Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i) leaves to the constructing module: Corollary 20.2 (ii), with Theorem 9.5 (ii) through Corollary 20.2 (iii) for an unconditioned receiver and the enclosure claim of the constructing submodule for a setting family a submodule constructs. On the exact route, with the port Grams enclosed exactly, every block $n \leq 1$ has radius zero (Theorem 9.9 (iii) for an unconditioned receiver, the exactness claim of the constructing submodule for a setting family a submodule constructs, and Corollary 20.2 (i)). The reference is read against the sector map of the floating-point blocks of (i) in the same entry (Definition "Sector maps and block-list operators" (foundations/def:sector-map)): the same block kind and block identifiers, the photon numbers $n = 0, \ldots, N$ in increasing order, and within each block the occupation vectors in the one order in which the entry lists the floating-point blocks, which is also the order of the rows and columns of every centre and radius matrix. The entries of the reference that a snap treats as known to be exactly zero (Definition "Snapping declared data onto a lattice" (foundations/def:snap)) are therefore exactly those enclosed as zero with radius zero.

(iv) Where the declaration has no model admissible exactly, where a declared R fails (D1) or (D2) exactly, or where a reciprocal that the

enclosure of a setting family a submodule constructs forms receives no enclosure, the entry carries no reference, records the offered reference kind of every block as *none*, and records the cause.

(v) R remains emitted in `dark_count_channel` (Requirement 22.8), where it serves the high-photon weight. It is already composed into the blocks of (i)–(iii), and a consumer does not compose it into them again.

(vi) Items (ii) and (iii) bind every run, save one that declares that enclosed references are not used (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i)). Such a run need form no enclosure of (ii) with a radius that is not zero, beyond what deciding that a block is not enclosed exactly takes, and carries none: it carries the reference of (iii) only on the blocks every entry of which it encloses exactly, with radius zero, from port Grams enclosed exactly, each block so carried being an exact reference of that block, and records their offered kind *exact*; every other block it records as offered with the kind *none*, with the cause that enclosed references are not used (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)). The reference the entry carries, its centres and radii as data, is the handed reference that the instance hands beside its snap of the emitted family (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap) (i)), whatever the snap itself uses of it: the snap uses only the centres of the exact blocks (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (ii)), and the run's first-order bound reads the reference on every block it carries (Requirement "The first-order bound evaluated on the reference companion" (meat-core/frank-wolfe/req:bracket-on-reference)). Item (iv) binds as stated.

(vii) *Factor enclosures.* Where the entry carries the reference of (iii) for a receiver with $\mathbb{I}_P = \varnothing$, whose one setting is off (Corollary 20.2 (iii)), the engine forms besides, on the blocks the count below admits and on no other, a factor enclosure (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)) of the block $\widetilde{F}^{(=n)}_{v|\mathrm{off}}$: the one of Theorem 9.7 (iii) at the weights $w_{S'} := R_{\varpi^{-1}(v)|S'}$, rationals by Corollary 20.2 (iv), at which the block of that theorem is $\widetilde{F}^{(=n)}_{v|\mathrm{off}}$ (Definition 20.1), its rows in the order of (iii), for the port dilation (Definition 9.6) formed as follows.

(a) On the route that declares a component graph, the dilation of Lemma 16.4 (ii), with the loss factor of part (i) of that lemma for every element. A row of a loss factor, and of the efficiency

layer's, is formed only where the declared value that scales it is decided in exact arithmetic to make it non-zero, so that whether an element is lossless is read off its declared parameters and never off a computed number. The loss factor of a generic block is formed from a decomposition $\mathbb{1}_\Sigma - G_\Sigma^\dagger G_\Sigma = E^\dagger D E$ formed and decided in exact rational arithmetic; where none is formed, no dilation of the declaration is.

(b) On a route that declares the port Grams or the instrument matrix, the dilation of Theorem 9.7 (iv), where the engine forms decompositions $A_k = E_k^\dagger D_k E_k$ and $\mathbb{L} = E_0^\dagger D_0 E_0$ of the kind named there, formed and decided in exact rational arithmetic.

The entries of the dilation are enclosed by the rules of Lemma 9.2 from square-root enclosures and angle enclosures, as Requirement 14.6 (iii) encloses the elements' matrices and the $\sqrt{\eta_k}$, and the columns as Theorem 9.7 (ii) encloses them, in exact rational arithmetic or by a rigorously enclosed computation (Definition "The tiers of a reported key rate" (foundations/def:tiers)), with no binary64 value as an input; a column enclosed exactly as zero is omitted, and the core is exact. The engine forms the factor enclosure of a block only where the number of occupation vectors $\boldsymbol{m}$ of weight n on the modes of the dilation whose row is not enclosed exactly as zero, with $R_{\varpi^{-1}(v)|S_{\boldsymbol{m}}} > 0$, is at most the block's dimension $\binom{|\mathbb{I}|+n-1}{n}$, a count it decides before it computes any column: a factor enclosure with more columns than that dimension fixes no rank (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (ii)). It forms none for a receiver with $\mathbb{I}_P \neq \varnothing$, whose blocks are not formed through a port dilation; none where no dilation of (a) or (b) is formed; none on a block the count does not admit; and none in a run that declares that enclosed references are not used (vi). A factor enclosure is carried beside the entrywise enclosure of its block, which it leaves unaltered, and is handed with the reference, altered by no step. The entry records per block whether one is carried, with its number of columns and its largest radius, or the cause where none is; and it names Theorem 9.7 and, on the route of (a), Lemma 16.4 as the claims that establish that the block is a member of its factor enclosure, which Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i) leaves to the module that constructs the reference.

Requirement 22.10 (Cause and fix recorded for a receiver block resting on its snap). When a moved block of a snap of the receiver's emitted dark-composed blocks (Requirement 22.9) rests on Assumption "The device

measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) under Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii), its record names, beside the setting, the block with its cause and the group that requirement names for it, the reference kind of the snap, as Requirement "Snapping by the declarer only" (foundations/req:who-snaps) defines it (*exact*, where the snap ran on the centres of an exact reference of the block, or *none*), the kind of the reference handed on the block, if any (*exact*, every radius of the block zero, or *enclosed*, with the largest radius of the block and the largest radius of the enclosures of irrational inputs (square roots, trigonometric values) from which the centres were computed exactly), and the fix below for the cause, which details the fix of its group. A run that declares the strict option of Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv) refuses instead, naming the same; no other run refuses for these causes, save where the last resort of Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap) (iv) yields no snap.

(i) *No reference.* The cause recorded by Requirement 22.9 (iv), in the group *no usable reference*, and the fix for it:

  (a) for declared port Grams that are not Hermitian, a declared family that is not an admissible Gram family exactly, or a generic block that is not a contraction exactly (Requirement 14.6 (i), (ii)), declared data that are Hermitian and admissible exactly, in rational arithmetic and not only at the receiver tolerances;

  (b) for a declared dark-count matrix that does not satisfy (D1) and (D2) exactly, one that is exactly column-stochastic and monotone, or the generated dark-count route;

  (c) for a reciprocal that the enclosure of a setting family a submodule constructs forms and that receives no enclosure, square-root and angle enclosures with smaller radii (Definition 9.1).

(ii) *Enclosed references not used.* Where the run declares that enclosed references are not used and the entry offers an enclosed one, the block is handed no reference, in the group *no usable reference*, and the fix names the kind the entry offers for every block (Requirement 22.9 (iii)), with the largest radius of every block offered as enclosed, and the alternative of an exact reference, which the receiver offers for a declaration on the exact route at a cutoff $N \leq 1$ when its port Grams are enclosed exactly (Theorem 9.9 (iii), Corollary 20.2 (ii)), or a run that uses enclosed references.

A block that the snap makes by the last resort of Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap) (iv) is

flagged `support_not_preserved` in the record, beside its cause and reference kind, and where it rests, in the group *last resort*, the fix names the ports whose declared efficiency is 1 where $\lambda_{\max}(A_\Sigma) = 1$, and the click patterns S whose entry $R_{S|\varnothing}$ of the dark-count channel is 0, since Proposition 20.3 (i) needs $R_{S|\varnothing} > 0$ for every S: on the declared dark-count route it asks for a declared R with $R_{S|\varnothing} > 0$ for every S, and on the generated dark-count route, where every $R_{S|\varnothing}$ is positive exactly when $0 < p_{\mathrm{dc}|k} < 1$ for every k (Proposition 20.3 (ii)), a dark-count probability in $(0, 1)$ in every port; in either case a model with loss, at the smallest loss and dark-count rates the user is sure the device has (Proposition 20.3); and, where the entry offers no reference, the fix of (i) as well. Where every element of a block of the reference is positive definite (Proposition 20.3 (i)), no row of it is structurally zero (Definition "Snapping declared data onto a lattice" (foundations/def:snap)), so every index lies in the structural support of every element and the uniform lift of that item meets the rules (i) to (iii) of Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap) on the block. For a block whose cause is the run's, a handed reference on which the companion's bound is withdrawn or not evaluated, or counts while the key length is formed from a number certified on the snapped declaration in its place, the cause `companion_not_certified` (Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (i)), the record of the run names the fix Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii) names for that cause, and, for such a block snapped by the last resort, which rests in the group *last resort*, that fix beside the one above; the run decides that cause after its search, so the fix is completed there and not at the snap. A handed block that the run evaluates off its snap rests on nothing and is named with no fix, whether or not the last resort snapped it. A reference that fails the decisions of Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i) cannot arise for the receiver: the centres are Hermitian with entries in $\mathbb{Q}(i)$ and the radii symmetric non-negative rationals by construction (Lemma 9.2 (v)), and the enclosure, positive semidefiniteness and completeness that the constructing module establishes are Corollary 20.2 (ii), whose hypothesis Theorem 9.5 (ii), through Corollary 20.2 (iii), and, for a setting family a submodule constructs, that submodule's enclosure of its blocks supply.

References